

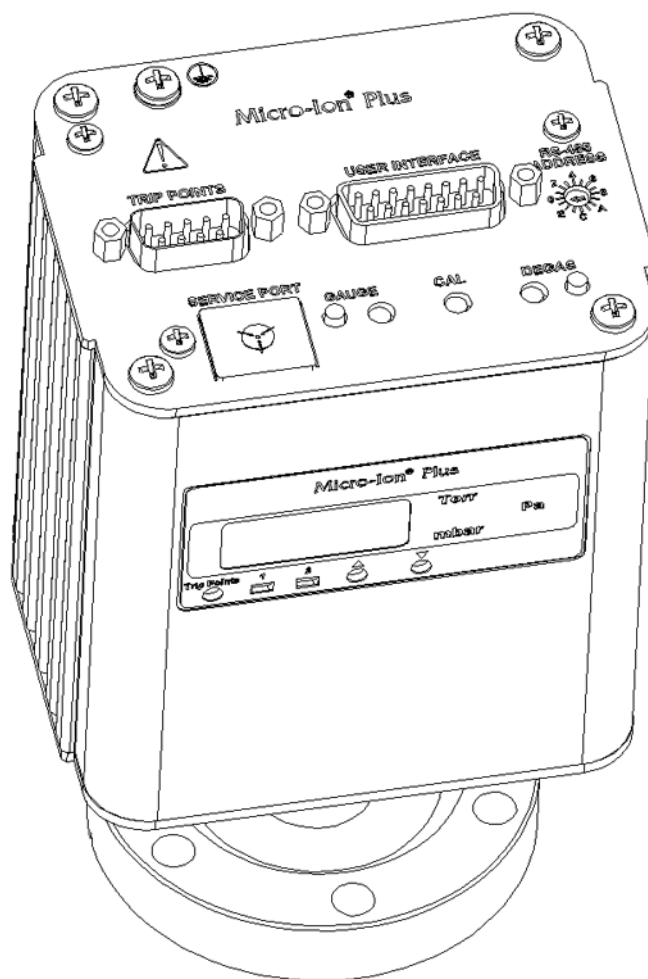
Series 356

Series 356 Micro-Ion[®] Plus Vacuum Gauge Module



Instruction Manual

*Instruction manual part number 356007-GP
Revision D - April 2018*



Series 356

Series 356 Micro-Ion[®] Plus Vacuum Gauge Module

This Instruction Manual is for use with all Granville-Phillips Series 356 Micro-Ion Plus Vacuum Gauge Modules. A list of applicable catalog numbers is provided on the following page.



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Instruction Manual

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Series 356 Micro-Ion[®] Plus Vacuum Gauge Module

Catalog numbers for Series 356 Micro-Ion Plus Vacuum Gauge Modules

Power supply and cable are not included.

- Module with RS-485 interface, analog output, no display:
- 356001 - Y # - #
- Module with RS-485 interface, analog output, digital display:
- 356002 - Y # - #

Ion Gauge Filaments:

Yttria-coated iridium

Y

Flange/Fitting:

- NW16KF
- NW25KF
- NW40KF
- 1.33 inch (NW16CF) ConFlat-type
- 2.75 inch (NW35CF) ConFlat-type
- 1/2 inch VCR-type male
- NW16L
- 40KFL

- D
- E
- K
- F
- G
- H
- M
- L

Measurement Units:

- Torr
- mbar
- pascal

- T
- M
- P

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Chapter 1 Introduction

1.1 About These Instructions

These instructions explain how to install, operate, and maintain the Granville-Phillips® Micro-Ion® Plus Vacuum Gauge Module. The module contains a Micro-Ion ionization gauge and a Conductron® heat-loss sensor, which working in combination provide pressure measurement from 1×10^{-9} to atmosphere.

- *This chapter* explains caution and warning statements, which must be adhered to at all times; explains your responsibility for reading and following all instructions; defines terms that are used throughout this manual; and tells you how to contact customer service.
- *Chapter 2* explains how to install the module.
- *Chapter 3* explains the theory of operation; module operating procedures; how to read the analog output; and how to use the RS-485 interface.
- *Chapter 4* explains troubleshooting; Micro-Ion gauge testing, removal and replacement; and module return procedures.
- *Appendix A* explains filament operation modes of the Micro-Ion Gauge.

1.2 Caution and Warning Statements

This manual contains caution and warning statements with which you *must* comply to prevent inaccurate measurement, property damage, or personal injury.



CAUTION

Caution statements alert you to hazards or unsafe practices that could result in minor personal injury or property damage.

Each caution statement explains what you must do to prevent or avoid the potential result of the specified hazard or unsafe practice.



WARNING

Warning statements alert you to hazards or unsafe practices that could result in severe property damage or personal injury due to electrical shock, fire, or explosion.

Each warning statement explains what you must do to prevent or avoid the potential result of the specified hazard or unsafe practice.

Caution and warning statements comply with American Institute of Standards Z535.1-2002 through Z535.5-2002, which set forth voluntary

practices regarding the content and appearance of safety signs, symbols, and labels.

Each caution or warning statement explains:

- a. The specific hazard that you *must* prevent or unsafe practice that you *must* avoid,
- b. The potential result of your failure to prevent the specified hazard or avoid the unsafe practice, and
- c. What you *must* do to prevent the specified hazardous result.

1.3 System Grounding

Grounding, though simple, is very important! Be certain that ground circuits are correctly used on your ion gauge power supplies, gauges, and vacuum chambers, regardless of their manufacturer. Safe operation of vacuum equipment requires grounding of all exposed conductors of the gauges, the controller and the vacuum system. **Lethal Voltages** may be established under some operating conditions unless correct grounding is provided.

Ion producing equipment, such as ionization gauges, mass spectrometers, sputtering systems, etc., from many manufacturers may, under some conditions, provide sufficient electrical conduction via a plasma to couple a high voltage electrode potential to the vacuum chamber. If exposed conductive parts of the gauge, controller, and chamber are not properly grounded, they may attain a potential near that of the high voltage electrode during this coupling. Potential fatal electrical shock could then occur because of the high voltage between these exposed conductors and ground.

1.4 Explosion / Implosion

Danger of injury to personnel and damage to equipment exists on all vacuum systems that incorporate gas sources or involve processes capable of pressuring the system above the limits it can safely withstand.

For example, danger of explosion in a vacuum system exists during backfilling from pressurized gas cylinders because many vacuum devices such as ionization gauge tubes, glass windows, glass belljars, etc., are not designed to be pressurized.

Install suitable devices that will limit the pressure from external gas sources to the level that the vacuum system can safely withstand. In addition, install suitable pressure relief valves or rupture disks that will release pressure at a level considerably below that pressure which the system can safely withstand.

Suppliers of pressure relief valves and pressure relief disks can be located via an online search, and are listed on ThomasNet.com under "Relief Valves" and "Rupture Discs. ***Confirm that these safety devices are properly installed before installing the Series 356 vacuum gauge module.***

In addition, check that (1) the proper gas cylinders are installed, (2) gas cylinder valve positions are correct on manual systems, and (3) the automation is correct on automated systems.

- 1.5 Operation**
- It is the installer's responsibility to ensure that the automatic signals provided by the process control module are always used in a safe manner. Carefully check manual operation of the system and the setpoint programming before switching to automatic operation. Where an equipment malfunction could cause a hazardous situation, always provide for fail-safe operation. As an example, in an automatic backfill operation where a malfunction might cause high internal pressures, provide an appropriate pressure relief device.
- 1.6 Reading and Following Instructions**
- You must comply with all instructions while you are installing, operating, or maintaining the module. Failure to comply with the instructions violates standards of design, manufacture, and intended use of the module. MKS Instruments, Inc. disclaims all liability for the customer's failure to comply with the instructions.
- *Read instructions* – Read all instructions before installing or operating the product.
 - *Retain instructions* – Retain the instructions for future reference.
 - *Follow instructions* – Follow all installation, operating and maintenance instructions.
 - *Heed warnings and cautions* – Adhere to all warnings and caution statements on the product and in these instructions.
 - *Parts and accessories* – Install only those replacement parts and accessories that are recommended by Granville-Phillips. Substitution of parts is hazardous.
- 1.7 Customer Service**
- If the product must be returned to the factory for service, request a Return Material Authorization (RMA) from MKS, which can be completed at <https://www.mksinst.com/service/servicehome.aspx>. Do not return products without first obtaining an RMA. In most cases a hazardous materials disclosure form is required. The MKS Customer Service Representative will advise you if the hazardous materials document is required.
- When returning products to MKS, be sure to package the products to prevent shipping damage. Shipping damage on returned products as a result of inadequate packaging is the Buyer's responsibility.
- For Customer Service / Technical Support:
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- 1.8 Definitions of Terms**
- Table 1-1 lists terms describing the Micro-Ion Plus Module and its components.

Table 1-1 Terms Describing the Micro-Ion Plus Module and Components

Term	Description
Module	The Micro-Ion Plus Module, which contains a hot filament Micro-Ion gauge (Bayard-Alpert type ionization gauge) and a Conductron heat-loss sensor.
Micro-Ion gauge	The Bayard-Alpert type ionization gauge, which indicates pressure by producing a current that is proportional to gas density.
Conductron sensor	The heat-loss sensor, which measures pressure as a function of heat loss through the gold-plated tungsten sensing wire.

1.9 Specifications

Measurement Range for Air and N₂

	Measurements will change with different gases and gas mixtures. Micro-Ion Plus gauges are not intended for use with flammable or explosive gases. Atmospheric pressure value is based on calibration at time of use.
Torr	1×10^{-9} Torr to atmosphere ("At")
mbar	1×10^{-9} mbar to atmosphere ("At")
pascal	1×10^{-7} pascal to atmosphere ("At")

Micro-Ion Gauge Emission Current

Emission current	0.1 and 4 mA, autoranging
Micro-Ion gauge auto on	2×10^{-2} Torr, 2.66×10^{-2} mbar, 2.66 pascal, with decreasing pressure
Micro-Ion gauge auto off	3×10^{-2} Torr, 3.99×10^{-2} mbar, 3.99 pascal, with increasing pressure
Micro-Ion gauge degas	Electron bombardment, 3.75 W with 2-minute timer
Default switch to high (4.0 mA)	5.0×10^{-6} Torr, 6.6×10^{-6} mbar, 6.6×10^{-4} pascal, with decreasing pressure
Default switch to low (0.1 mA)	1×10^{-5} Torr, 1.3×10^{-5} mbar, 1.3×10^{-3} pascal, with increasing pressure, adjustable, 50% hysteresis (automatically set)

Filament selection	Alternating, automatic, or manual modes For a Micro-Ion gauge with yttria-coated iridium filaments, the default is alternating
---------------------------	--

Outputs and Indicators

Analog output	Logarithmic, 0.50 Vdc/decade, where 1000 Torr = 7 Vdc
Digital RS-485 output	
<i>Interface</i>	RS-485 two-wire, half-duplex
<i>Communications format</i>	ASCII format, eight data bits, no parity, one stop bit
<i>Baud rates</i>	1200, 2400, 4800, 9600, 19200 (default), or 38400 baud
<i>Address</i>	0 to 63, selected by using address switch and RS-485 SA command
<i>LED status indicator</i>	Module status indicator lights up to indicate module status.
Pressure display (optional)	X.X±Y display format includes two significant digits, a 1-digit exponent, and a ± sign for the exponent.
Trip point indicators (optional)	LED indicator for relay 1 or relay 2 is solid green when corresponding relay is activated or blinks green when corresponding relay is being configured.

Trip point relays

Relay type	Two single-pole, double-throw relays (SPDT) Relay contacts are silver alloy-gold clad, rated for 1 A at 30 Vdc. The relays can handle resistive or non-inductive loads.
Relay contact ratings	1 A at 30 Vdc resistive or non-inductive
<i>Minimum hysteresis</i>	5%
<i>Range</i>	1.0 × 10 ⁻⁹ to 100 Torr 1.0 × 10 ⁻⁹ to 133 mbar 1.0 × 10 ⁻⁷ to 1.33 × 10 ⁴ pascal
Default activation pressure	Default activation pressures are set to 0.0.

Micro-Ion Plus Module

Micro-Ion gauge sensitivity	
<i>At 4 mA emission (high)</i>	18/Torr, 13.5/mbar, .135/pascal
<i>At 0.1 mA emission (low)</i>	20/Torr, 15/mbar, .15/pascal (± 15% variation)
X-ray limit	< 3 × 10 ⁻¹⁰ Torr, < 3.99 × 10 ⁻¹⁰ mbar, < 3.99 × 10 ⁻⁸ pascal The x-ray limit is the absolute lowest indication from the gauge. It is not

Micro-Ion gauge filament material	practical to make repeatable measurements near the x ray limit Yttria-coated iridium
Conductron sensing wire material	Gold-plated tungsten
Internal volume	10.8 cm ³ , (0.67 in. ³) to the port screen

Physical Characteristics

Weight	567 gm (20 oz.) with NW16KF fitting
Power required	+24 Vdc ±15% external power supply. Power supply must provide 1.5 A current at 24 Vdc. Inrush current can momentarily exceed the 1.5 A peak. +24 Vdc±15% external power supply must be certified to IEC standard with a safety extra-low voltage classified output.
Operating environment	+10 °C to +40 °C ambient, non-condensing, indoor use only, ordinary protection for moisture, maximum altitude 3000 meters
Operating conditions	Suitable for continuous operation. Category 1 for installation over voltage. Pollution degree 2, Class 1
Non-operating temperature	−40 °C to +85 °C (−40 °F to +185 °F)
Case material	Aluminum extrusion
Connectors	15-pin subminiature D connector for module power supply, analog outputs, and RS-485 outputs
Materials exposed to gas	Yttria-coated iridium, gold-plated tungsten, 304 stainless steel, tantalum, tungsten, nickel iron alloy, nickel, borosilicate glass
Compliance	
<i>Product Safety</i>	IEC/EN 61010-1
<i>IP Rating</i>	IP20

2.1 Module Components

The Micro-Ion Plus Module contains a Micro-Ion gauge (Bayard-Alpert type ionization gauge) and a Conductron heat-loss sensor.



WARNING

Using the module to measure the pressure of flammable or explosive gases can cause a fire or explosion resulting in severe property damage or personal injury.

Do not use the module to measure the pressure of flammable or explosive gases.



WARNING

Exposing the module to moisture can cause fire or electrical shock resulting in severe property damage or personal injury.

To avoid exposing the module to moisture, install the module in an indoor environment. Do not install the module in any outdoor environment.

2.2 Installation Procedure

The module installation procedure includes the following steps:

1. Install the appropriate pressure relief devices in the vacuum system.
2. Determine the best for the module on the vacuum chamber.
3. Attach the module vacuum chamber fitting to its mate on the vacuum chamber.
4. Assemble and connect the module wiring.
5. Calibrate the Conductron sensor at atmosphere.

This chapter also explains what to do if radio frequency interference (RFI) from the module disrupts operation of other electronic devices.

Step 1 *Install Pressure Relief Devices*

Before you install the module, install appropriate pressure relief devices in the vacuum system.

MKS Instruments, Inc./Granville-Phillips does not supply pressure relief valves or rupture disks. Suppliers of pressure relief valves and pressure relief disks can be located via an online search, and are listed on ThomasNet.com under “Relief Valves” and “Rupture Discs. ***Confirm that these safety devices are properly installed before installing and operating the product.***

Ensure the following precautions are complied with at all times:

- (1) the proper gas cylinders are installed,
- (2) the gas cylinder valve positions are correct on manual systems,
- (3) and the automation is correct on automated gas delivery systems.



CAUTION

Operating the module above 1000 Torr (1333 mbar, 133 kPa) true pressure could cause vacuum system rupture or product failure.

To avoid vacuum system rupture or product failure due to overpressurization, install pressure relief valves or rupture disks in the system if pressure exceeds 1000 Torr (1333 mbar, 133 kPa).

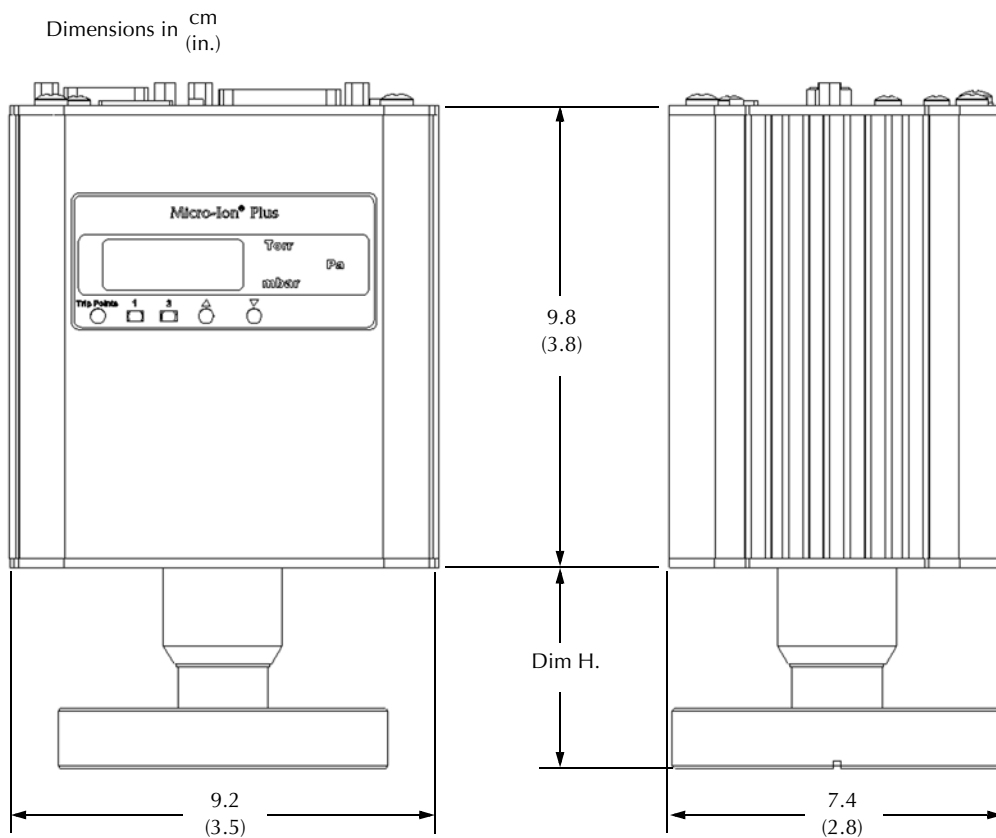
Step 2 *Location of the Module*

To determine the best location on the vacuum system for module, refer to Figure 2-1 and Table 2-1, and the guidelines below.

- For proper operation of the module at atmosphere, orient the module vertically (see Figure 2-1). Although the Micro-Ion gauge will read correctly below 100 Torr with the module mounted in any position, inaccurate readings will result at pressures above 100 Torr if the module is not in the vertical position as illustrated in Figure 2-1.
- For greatest accuracy and repeatability, locate the module in a stable, room-temperature environment. Ambient temperature should never exceed 40 °C (104 °F) operating, non-condensing, or 85 °C (185 °F) non-operating.
- Locate the module away from internal and external heat sources and in an area where ambient temperature remains reasonably constant.
- Do not locate the module near the pump, where gauge pressure might be lower than normal vacuum chamber pressure.

- Do not locate the module near a gas inlet or other source of contamination, where inflow of gas or particulates causes atmospheric pressure to be higher than system atmosphere.
- Do not locate the module where it will be exposed to corrosive gases such as mercury vapor or fluorine.

Figure 2-1 Dimensions



Step 3 *Attach the Module to the Vacuum Chamber*

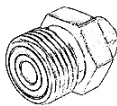
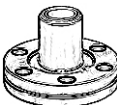
Attach the module vacuum chamber fitting to its mate on the vacuum chamber.

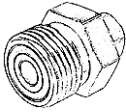
**CAUTION**

Twisting the module to tighten the fitting to the vacuum chamber can damage the module's internal connections.

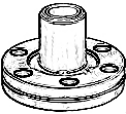
- **Do not twist the module to tighten the fitting.**
- **Use appropriate tools to tighten the fitting.**

Table 2-1 **Fittings for the Micro-Ion Plus Module**

Fitting		Dim. H	
		cm	in.
	1/2-inch VCR-type male	5.8	2.3
	1.33-inch (NW16CF) ConFlat type	4.3	1.7
	2.75-inch (NW35CF) ConFlat type	4.3	1.7
	NW16KF	2.0	0.8
	NW25KF	2.0	0.8
	NW40KF	2.0	0.8
	NW16KF extended	4.2	1.65
	NW25KF extended	4.2	1.65
	NW40KF extended	4.6	1.8

VCR-type fitting

- Remove the plastic or metal bead protector cap from the fitting.
- If a gasket is used, place the gasket into the female nut.
- Assemble the components and tighten them finger-tight.
- While holding a back-up wrench stationary, tighten the female nut 1/8 turn past finger-tight on 316 stainless steel or nickel gaskets, or 1/4 turn past finger-tight on copper or aluminum gaskets. *Do not twist the module to tighten the fitting.*

ConFlat flange

To minimize the possibility of leaks with ConFlat flanges, use high strength stainless steel bolts and a new, clean stainless steel with OFHC copper gasket. Avoid scratching the seal surfaces.

- Finger tighten all bolts.
- Use a wrench to continue tightening 1/8 turn at a time in crisscross order (1, 4, 2, 5, 3, 6) until flange faces make contact. Further tighten each bolt about 1/16 turn.

NW flange

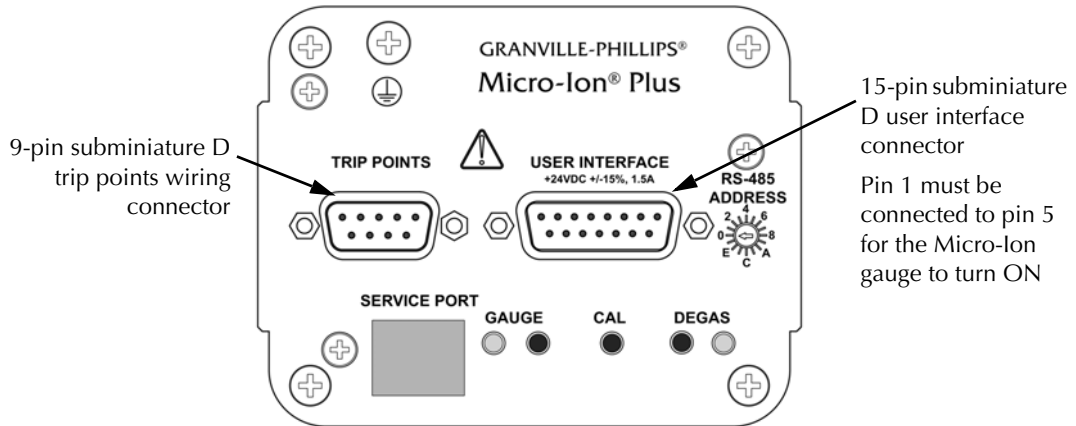
The NW mounting system requires O-rings and centering rings between mating flanges.

Tighten the clamp to compress the mating flanges together and seal the O-ring.

Step 4 *Assemble and Connect the Cables***Connecting Cables**

The cables are user-supplied. Granville-Phillips does not supply cable.

- Use externally shielded cable.
- Connect the wiring to the 9-pin trip point connector and the 15-pin subminiature D connector. On the User Interface connector, Pin 1 must be connected to pin 5 for the Micro-Ion gauge to turn ON. See Figure 2-2.

Figure 2-2 9-pin Trip Point and 15-pin User Interface Subminiature D Wiring Connectors**Module Power Supply**

Connect the module power supply to terminals 5 and 8 on the 15-pin I/O wiring connector.

- Pin 5 (ground) is negative (–).
- Pin 8 (power input) is positive (+).

The module requires a +24 Vdc $\pm 15\%$ external power supply. The power supply must provide a 1.5 A current at 24 Vdc. Inrush current can momentarily exceed the 1.5 A peak.

Typical module operating power is 18 W for 4 mA emission when the Micro-Ion gauge is ON.

Power inputs are reverse-bias protected.

The Micro-Ion gauge will not activate and an emission error will occur if insufficient power is supplied during Micro-Ion gauge activation.

Analog and RS-485 Output Wiring

Connect wiring for analog and RS-485 outputs to the 15-pin subminiature-D connector, as illustrated in Figure 2-3.

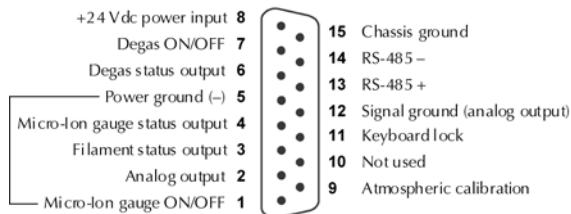
Figure 2-3 Wiring Terminals for Analog and RS-485 Wiring

Table 2-2 Connections to 15-pin Subminiature D Connector

Pin #	Function	Input or Output	Description
Pin 1	Micro-Ion gauge ON/OFF	Input	Must be continuously connected to Pin 5 (LOW) to turn gauge ON (a) • Removing ground connection turns gauge OFF
Pin 2	Analog output	Output	• 0.5 Vdc/decade, 7 Vdc maximum • Ground reference to pin 12
Pin 3	Filament status	Output	(LOW) Indicates an inoperable filament (b)
Pin 4	Gauge status	Output	(LOW) Indicates Micro-Ion gauge is ON (b)
Pin 5	Ground (–)	Input	Power supply ground
Pin 6	Degas status	Output	(LOW) Indicates Micro-Ion gauge degas is ON (b)
Pin 7	Degas ON/OFF	Input	• Continuous LOW will activate 2-minute degas (a) • Pressure must be $< 5 \times 10^{-6}$ Torr (6.66×10^{-6} mbar, 6.66×10^{-4} pascal) for degas to start
Pin 8	+24 Vdc power	Input	+20.4 Vdc to +27.6 Vdc, 26 W nominal
Pin 9	Calibration	Input	LOW sets atmospheric or vacuum calibration point (a) (c)
Pin 10	Not used		Not used
Pin 11	Keyboard lock	Input	Continuous LOW locks keyboard (a)
Pin 12	Signal common	Output	Signal ground (analog output)
Pin 13	RS-485+	Input/output	RS-485+
Pin 14	RS-485 –	Input/output	RS-485 –
Pin 15	Chassis ground	Input	Chassis ground (cable shield)

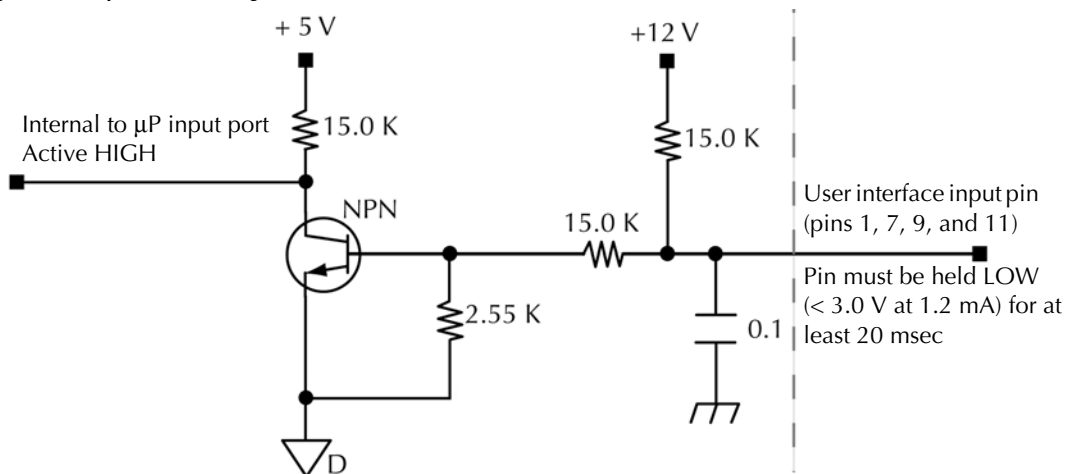
(a) 15K pull-up to 12 Vdc, (LOW) 0 Vdc at 0.80 mA. See Figure 2-4.

(b) Open Collector, no pull-up, 40 Vdc, 50 mA. See Figure 2-4.

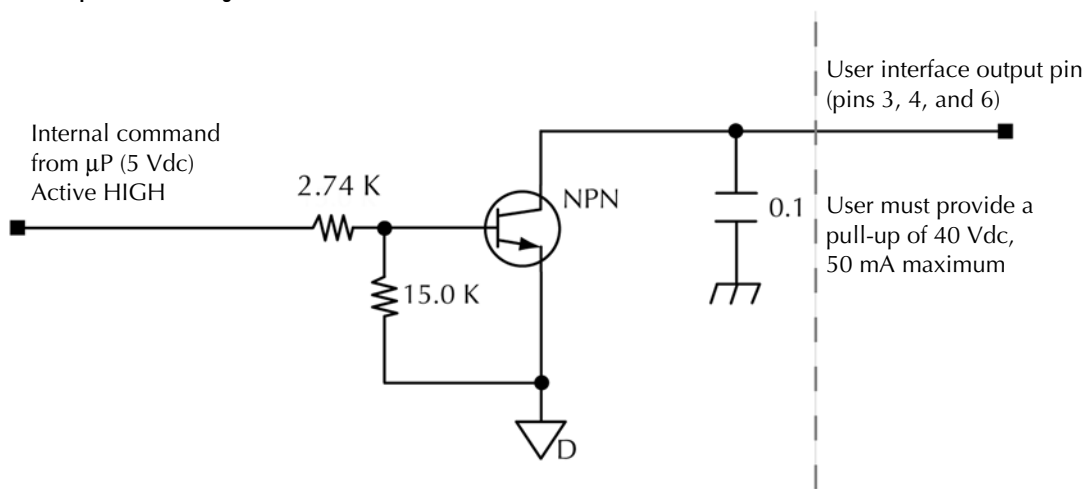
(c) Pin must be held LOW for at least 2 msec to activate calibration.

Control Inputs

The module provides an internal 15 K pull-up resistor to 12 Vdc, as illustrated in Figure 2-4.

Figure 2-4 Input Circuit Wiring**Status Outputs**

The module provides an open collector output with no pull-up, as illustrated in Figure 2-5.

Figure 2-5 Output Circuit Wiring

Relay Wiring

The module has two trip point relays. The contacts are silver alloy-gold clad, rated for 1 A at 30 Vdc. The relays can handle resistive or non-inductive loads.

- Figure 2-6 illustrates the 9-pin trip point connector.
- Table 2-3 lists pin connections for the 9-pin male D connector.

Figure 2-6 Trip Point Relay 9-pin Connector

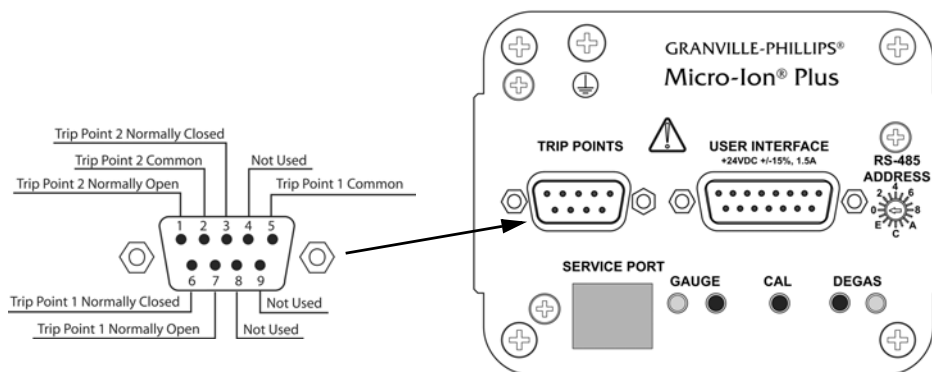


Table 2-3 Trip Point Relay 9-pin Connector Pins

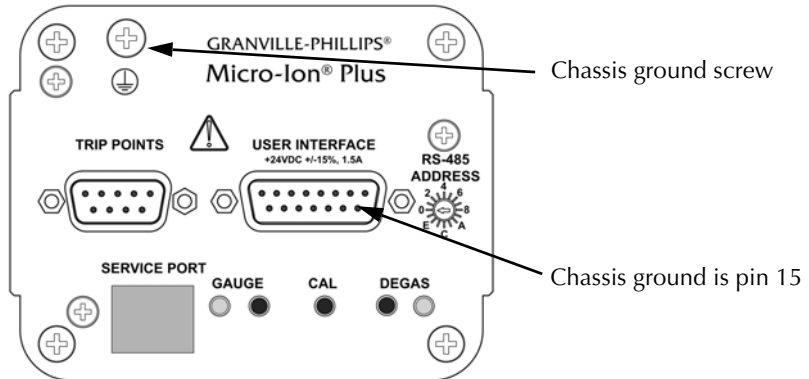
Pin #	Function	Input/Output	Voltage Level
<i>These voltage levels apply ONLY for resistive or non-inductive loads.</i>			
Pin 1	Relay 2: Normally open	Output	0 to 30 Vdc, 1 A
Pin 2	Relay 2: Common	Output	0 to 30 Vdc, 1 A
Pin 3	Relay 2: Normally closed	Output	0 to 30 Vdc, 1 A
Pin 4	Not used		No connection
Pin 5	Relay 1: Common	Output	0 to 30 Vdc, 1 A
Pin 6	Relay 1: Normally closed	Output	0 to 30 Vdc, 1 A
Pin 7	Relay 1: Normally open	Output	0 to 30 Vdc, 1 A
Pin 8	Not used		No connection
Pin 9	Not used		No connection

Grounding

The module chassis must be properly grounded via the 15-pin subminiature D connector and the chassis ground screw, as illustrated in Figure 2-7.

The module generates 180 Vdc during normal operation (Micro-Ion gauge ON) and 250 Vdc during Micro-Ion gauge degas.

Figure 2-7 Ground Connections

**WARNING**

Improper grounding could cause product failure or personal injury.

Follow ground network requirements for the facility.

- Maintain all exposed conductors at earth ground.
- Make sure the vacuum port to which the module is mounted is properly grounded.
- Connect the ground strap to the chassis ground screw on the module end panel.

Step 5 *Calibrate the Conductron Sensor*

To calibrate the Conductron sensor, see *Calibrate the Module at Atmosphere* on page 35.

An atmospheric calibration is performed on the Conductron sensor at the factory before the module is shipped. The factory calibration sets the atmospheric calibration point to approximately 635 Torr (846 mbar, 84.6 kPa).

Because atmospheric pressure varies according to location, you should set the atmospheric calibration point, *using N₂ or air*, before putting the module into operation. Periodic resets of the atmospheric calibration point also improve the accuracy and repeatability of the Conductron sensor.

2.3 **Eliminating Radio Frequency Interference**

The module complies with U.S. Federal Communications Commission (FCC) limits for a Class A digital device, pursuant to Part 15 of the FCC rules. These limits provide reasonable protection against harmful interference when the module operates in a commercial environment.

The module generates and can radiate radio frequency energy and, if not installed and used in accordance with the instructions in this manual, may cause harmful interference to radio and television communications. However, there is no guarantee that interference will not occur in a particular installation. If the module causes interference to radio or television reception, which can be determined by turning the module OFF and ON, use one of the following methods to eliminate the interference:

- Re-orient or relocate the receiving antenna.
- Increase the separation between the module and the receiver.
- Connect the module into an outlet on a circuit that is *not* the circuit to which the receiver is connected.
- Consult an experienced radio or television technician for help.

3.1 Preparing to Operate the Module

This chapter explains how to operate the Series 356 Micro-Ion Plus Module.



WARNING

Do not use the module to measure the pressure of flammable or explosive gases. Using the module to measure the pressure of flammable or explosive gases can cause a fire or explosion resulting in severe property damage, personal injury, or death.

The module can be operated by providing power to the module as described in the Installation chapter and using the analog output signal, or the RS-485 interface, or by using the optional display.

Before putting the module into operation, you must perform the following procedures:

1. Install the module in accordance with the instructions on pages 13–23.
2. Connect Pin 1 (Micro-Ion Gauge ON/OFF) of the User Interface Connector to ground (Pin 5), or it must be controlled by the user's process control.
3. Develop a logic diagram of the process control function.
4. Develop a diagram detailing the Analog Output connections.
5. Develop a diagram of the connections to the RS-485, I/O status, and control lines of the 15-pin User Interface connector.
6. Develop a circuit schematic that specifies exactly how each piece of system hardware will connect to the module relays.
7. Attach a copy of the process control circuit diagram to this manual for future reference and troubleshooting.
8. For units with the RS-485 interface, set the module address, baud rate, and parity as instructed on pages 41 – 42. All RS-485 command instructions are listed in the section titled *RS-485 Operation and Commands* beginning on page 40.

If you need application assistance, phone a Granville-Phillips application engineer at **1-303-652-4400** or email mks@mksinst.com.

3.2 Theory of Operation

The Micro-Ion Plus vacuum gauge module consists of two separate pressure measuring devices: a hot filament Micro-Ion gauge (Bayard-Alpert type ionization gauge), and a Conduction heat-loss sensor. When power is applied to the module, the Conduction sensor is ON. As the system is pumped down, the Conduction sensor turns ON the Micro-Ion gauge at the pressure switch point. As pressure increases, the Micro-Ion gauge is turned OFF by the Conduction sensor. The measurement range of the Micro-Ion gauge and the Conduction sensor overlap. When pressure is within the measurement range of the gauge and the sensor, the pressure output is a blended signal in the “transition range” between the gauge and the sensor.

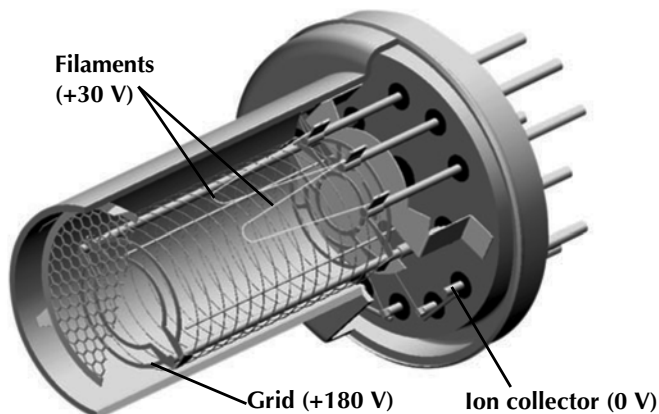
Micro-Ion Gauge Operation

The functional parts of the Micro-Ion gauge are the filaments (cathodes), grid (anode) and an ion collector, as illustrated in Figure 3-1. These electrodes are maintained by the controller at +30, +180, and 0 volts, relative to ground, respectively.

The filaments are heated to such a temperature that electrons are emitted and accelerated toward the grid by the potential difference between the grid and filaments. All of the electrons eventually collide with the grid, but many first traverse the region inside the grid many times.

When an electron collides with a gas molecule, an electron is dislodged from the molecule, leaving it with a positive charge. Most ions are then accelerated to the ion collectors. The rate at which electron collisions with molecules occur is proportional to the density of gas molecules, and hence the ion current is proportional to the gas density (or pressure, at constant temperature).

Figure 3-1 Micro-Ion Gauge



The amount of ion current for a given emission current and pressure depends on the Micro-Ion gauge design. This gives rise to the definition of ionization gauge sensitivity, frequently denoted by “S”:

$$S = \frac{\text{Ion current}}{\text{Emission current} \times \text{Pressure}}$$

When used with N₂ or air, the ionization gauge has a nominal sensitivity of 18/Torr (13.5/mbar, .135/pascal) at high emission current (4 mA) or 20/Torr (15/mbar, .15/pascal) at low emission current (0.1 mA).

The module electronics for the gauge varies the heating current to the filament to maintain a constant electron emission and measures the ion current to the ion collectors. The pressure is then calculated from these data.

Micro-Ion gauge degas is accomplished by increasing the emission current to 15 mA and raising the grid bias to 250 Vdc, resulting in an increased temperature of the grid to drive off adsorbed gases.

Conductron Sensor Operation

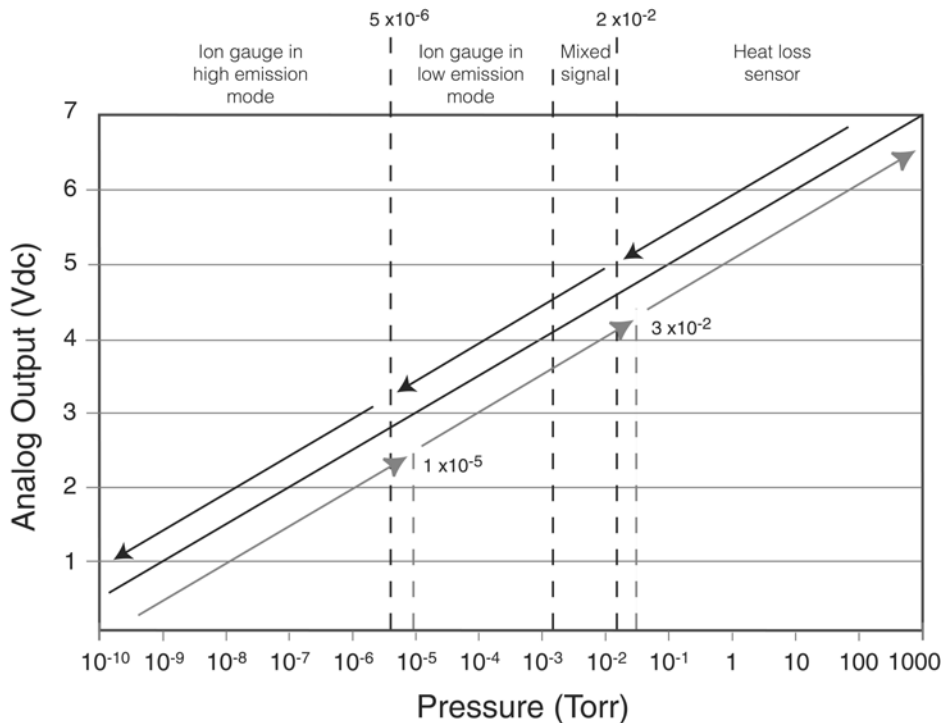
The Conductron heat-loss sensor uses Granville-Phillips’ Conductron sensor proprietary geometry and control circuitry. The sensor is comprised of two coplanar wire elements, a sensing wire, and a compensating wire that corrects for ambient conditions. The resistance of the sensing wire increases as its temperature increases. The controller continually adjusts a heating current that flows through the sensing wire to keep the sensor at a regulated temperature.

If pressure increases, the heat loss through gas conduction from the sensor increases, and the sensor temperature decreases, causing the resistance of the sensor to decrease. As this occurs, an error amplifier senses a change in the resistance differential between the sensor element and the compensation element, and generates an increase in the heating current through the sensor. The increased current through the sensor increases its temperature (and its resistance), and the resistance differential between the two elements is reestablished at a higher sensor input voltage and current. When calibrated, using a series of pressures and ambient temperatures, the input voltage and current are measured as an indication of the system pressure.

Ion Gauge/Conductron Gauge Auto On/Off

As the vacuum system pumps down from atmosphere, the Conductron sensor (heat loss sensor) measures pressure until a sufficiently low pressure level is achieved, then automatically turns ON the Micro-Ion gauge in the low emission mode. During the first decade that both sensors are operating, the control electronics mixes the signals. As the pressure is further reduced, the Micro Ion gauge switches from low emission to high emission. As pressure increases, the opposite occurs at slightly higher pressures. This sequence is illustrated in Figure 3-2.

Figure 3-2 Auto Ranging Actuation Points



Other emission and filament operational modes are available to meet specific application requirements. Contact a Customer Service Engineer for help with your particular application. Contact information is provided on page 3 and on the back cover of this instruction manual, and *Customer Service* on page 59.

3.3 Set the Micro-Ion Gauge Emission Current Threshold

The Micro-Ion gauge can operate at either of two emission current levels.

- In low-emission mode, the current level is 0.1 mA.
- In high-emission mode, the current level is 4 mA.

As the vacuum pump removes gas from the system, the Conductron sensor measures pressure until it has decreased to a threshold pressure at which the Micro-Ion gauge can operate. At this gauge pressure, the Conductron sensor turns ON the Micro-Ion gauge at the low emission current level (0.1 mA).

As pressure continues to decrease, the Micro-Ion gauge switches from low emission to high emission (4 mA). If pressure increases after the emission current has gone from low to high, the gauge switches back to low emission. Table 3-1 lists default, minimum, and maximum pressure values at which the gauge switches emission current.

Table 3-1 Micro-Ion Gauge Emission Current Pressure Values

Emission Current Setting	Minimum Switch Points	Default Switch Points	Maximum Switch Points
Switch to high emission current (4 mA) with decreasing pressure	5.0×10^{-8} Torr 6.6×10^{-8} mbar 6.6×10^{-6} pascal	5.0×10^{-6} Torr 6.6×10^{-6} mbar 6.6×10^{-4} pascal	5.0×10^{-5} Torr 6.6×10^{-5} mbar 6.6×10^{-3} pascal
Switch to low emission current (0.1 mA) with increasing pressure	1.0×10^{-7} Torr 1.3×10^{-7} mbar 1.3×10^{-5} pascal	1×10^{-5} Torr 1.3×10^{-5} mbar 1.3×10^{-3} pascal	1.0×10^{-4} Torr 1.3×10^{-4} mbar 1.3×10^{-2} pascal

The switch back to low emission current with increasing pressure is 100% greater than the switch to high emission current with decreasing pressure.

For example, in default mode, the emission current switches from low to high emission at 5×10^{-6} Torr (6.6×10^{-6} mbar, 6.6×10^{-4} pascal), then switches back to low emission at 1×10^{-5} Torr (1.3×10^{-5} mbar, 1.3×10^{-3} pascal).

The emission switching pressures may need to be changed to adapt to the pressures of your process.

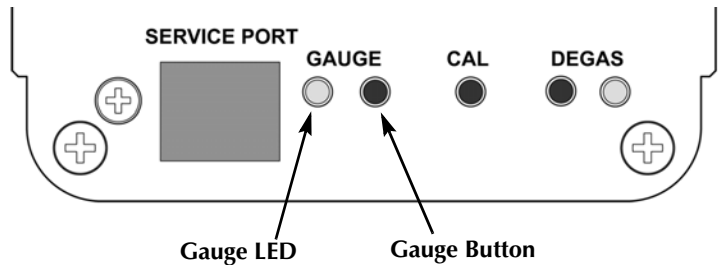
See *SER - Set Emission Range* on page 52 to set or read the pressure at which the Micro-Ion gauge switches from high to low emission current with increasing pressure.

3.4 Set the Micro-Ion Gauge Filament Mode

The Micro-Ion gauge contains two filaments. The Micro-Ion gauge can operate in automatic, alternating, or manual filament mode. Appendix A contains tables that explain the operation of the Micro-Ion gauge, its filaments, the Gauge LED, and the Gauge button in automatic, alternating, and manual modes. Figure 3-3 illustrates the Gauge button and Gauge LED on the module control panel.

For RS-485 operation, see *SF - Set Filament* on page 52.

Figure 3-3 Gauge LED



Automatic Filament Selection

As the vacuum system pumps down from atmosphere, the Conductron sensor measures pressure until a pressure of 20 mTorr (2.66×10^{-2} mbar, 2.66 pascal) is achieved, then automatically turns ON the Micro-Ion gauge. Tungsten filaments are more likely than yttria-coated iridium filaments to burn out if they turn ON at a pressure that is too high. To reduce the risk of burnout, the default behavior of the Micro-Ion gauge depends on the filament material.

If a rapid increase in pressure from high vacuum levels to pressures of 1 Torr (1.33 mbar, 133 pascal) or higher pressure occurs, tungsten filaments are almost certain to burn out. This risk is not unique to the Micro-Ion gauge and exists for all ion gauges with tungsten filaments.

The catalog number on the Micro-Ion Plus Module will indicate whether your module has yttria-coated iridium filaments or tungsten filaments.

- 356xxx -Y-x-x : Y = Yttria coated iridium
- 356xxx-T-x-x : T = Tungsten

Table 3-2 lists default delay times and filament modes for the Micro-Ion gauge.

Using RS-485 communications, you can use the IDT and SF commands to change the behavior of the Micro-Ion gauge. See page 51.

Table 3-2 Defaults for Micro-Ion Gauge Filament Material

Function	Default for Tungsten Filaments	Default for Yttria-coated Iridium Filaments
Micro-Ion gauge delay time	Module software determines the appropriate amount of time to wait before the Micro-Ion gauge turns ON with decreasing pressure	Micro-Ion gauge turns ON, without delay, as soon as operational pressure is achieved with decreasing pressure.
Micro-Ion gauge filament mode	Manual (see Table 3-17).	Alternating (see Table 3-17).

3.5 Turn the Micro-Ion Gauge ON or OFF

When the module starts up, the Conductron sensor is ON. As pressure decreases to a level that allows the Micro-Ion gauge to operate, the sensor turns ON the Micro-Ion gauge.

If the Micro-Ion gauge is in manual filament mode and one filament is inoperable, you must switch to the other filament by sending the IG0 command, then send the IG1 command, or push the Gauge button.

You can also use the module gauge button to turn the Micro-Ion gauge ON or OFF. Figure 3-4 illustrates the gauge button and gauge LED on the module control panel.

You can also install a switch between pins 1 and 5 on the 15-pin subminiature D connector that enables the Micro-Ion gauge to be turned ON or OFF. Pin 1 must be grounded to pin 5 to enable the Micro-Ion gauge to operate. A process relay or switch can be used to perform this function.

- To turn ON the Micro-Ion gauge, the process relay will close the switch.
- To turn OFF the Micro-Ion gauge, the process relay will open the switch.

If the gauge is manually turned OFF, the optional numeric display on the module indicates pressure as measured by the Conductron sensor and the gauge status indicator turns to blinking amber.

Table 3-3 describes the gauge status and corresponding LED states.

Using RS-485 communications, you can use the IG and IGM commands to turn ON/OFF the Micro-Ion gauge. See page 50.

Figure 3-4 Micro-Ion Gauge Button and LED on Module Control Panel

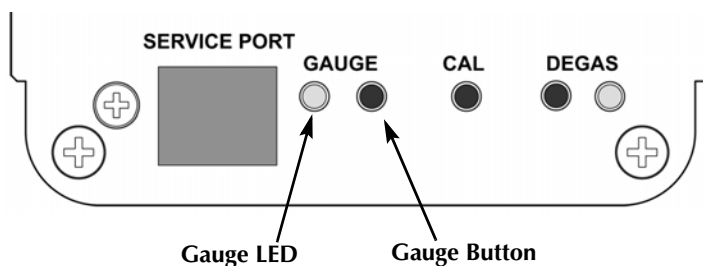


Table 3-3 Micro-Ion Gauge Status LED

Gauge Status	LED State
Micro-Ion gauge and Conductron sensor ON, both gauge filaments OK	Solid green
Micro-Ion gauge and Conductron sensor ON, one gauge filament cannot sustain emission	Blinking green
Micro-Ion gauge is OFF, pressure > turn-ON point	Solid amber
• Operator used switch or command to switch Micro-Ion gauge sensor OFF • Optional numeric display indicates pressure as measured by Conductron sensor	Blinking amber

Table 3-3 Micro-Ion Gauge Status LED

Gauge Status	LED State
• Micro-Ion gauge is OFF, both filaments are open • Conduction sensor failure • Micro-Ion gauge will not turn ON	Solid red
• Micro-Ion gauge is OFF, primary filament failure • if operating in manual filament mode, press gauge button to turn Micro-Ion gauge ON	Blinking red

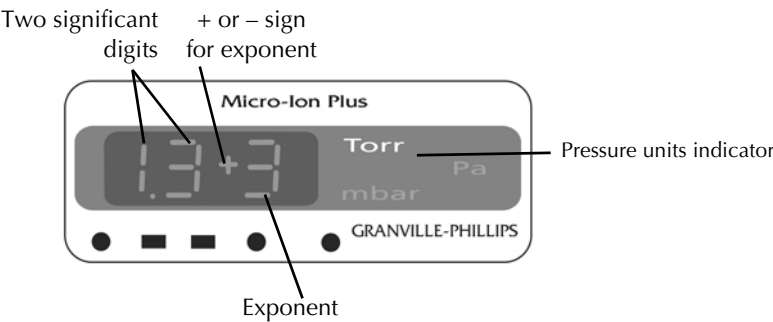
3.6 Read Vacuum Pressure

The optional numerical display is a 7-segment, green LED display. See Figure 3-5. The display shows 2 significant digits, 1-digit exponent, and a $\pm$ sign for the exponent, allowing for an $XX\pm Y$ display.

The displayed pressure range is 0.1×10^{-9} to atmosphere (“At”) as measured by the Conduction sensor and Micro-Ion gauge.

If the display reads “At+,” pressure is higher than 760 Torr (1000 mbar, 1.00×10^5 pascal), and calibration at atmosphere is required. See *Calibrate the Module at Atmosphere* on page 35.

Figure 3-5 Optional Numeric Display



Pressure values in the 10^{-10} range (such as 6×10^{-10}) will be displayed as “0.6–9”. The resolution of the numeric display changes over the operating range of the module. See Table 3-4.

Pressure will be displayed in units of Torr, mbar or pascal, as selected. The unit of measure is illuminated on the display panel. The units indicator is always ON when the module is powered.

Table 3-4 Display Resolution vs. Conductron Sensor Pressure

Measured Pressure			Display Resolution
Torr	mbar	Pascal	
Less than 50	Less than 66	Less than 6.6×10^3	Two digits
50 to 100	66 to 133	6.6×10^3 to 13.3×10^4	Increments of 10 pressure units
100 to 400	133 to 533	1.33×10^4 to 53.3×10^4	Increments of 100 pressure units
400 to 760	533 to 1.00×10^3	5.33×10^4 to 1.00×10^5	Display reads "At"
Greater than 760	Greater than 1.00×10^3	Greater than 1.00×10^5	Display reads "At+"

3.7 Read the Analog Output

- The output is logarithmic, 0.50 Vdc/decade, where 1000 Torr = 7 Vdc, as illustrated in Figure 3-6.
- Output impedance is 100 Ω .

An output signal of 10 Vdc occurs any time the gauge is intentionally turned OFF or when an error condition occurs.

Use the following equation to convert output voltage to pressure in Torr:

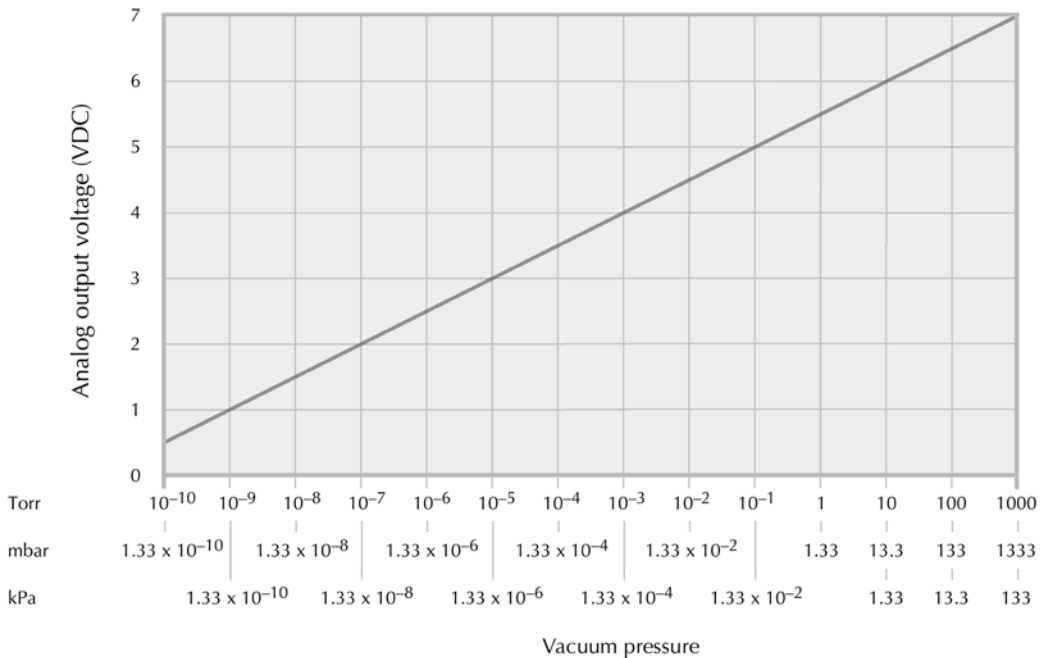
$$P_{\text{Torr}} = 10^{\left(\frac{V_{\text{out}} - 5.500}{0.500}\right)}$$

Use the following equation to calculate output voltage from pressure:

$$V_{\text{out}} = 0.500 \log(P_{\text{Torr}}) + 5.500$$

To convert Torr to mbar, multiply P_{Torr} by 1.333.

To convert Torr to pascal, multiply P_{Torr} by 133.3.

Figure 3-6 Analog Output Voltage vs. Pressure

3.8 Degas the Micro-Ion Gauge

You can use the module degas button to initiate a Micro-Ion gauge degas cycle. Press the degas button to initiate or terminate the degas cycle. The degas cycle will run for two minutes unless you terminate it earlier. The degas LED turns solid amber to indicate degas in progress. Figure 3-7 illustrates the degas button and LED on the module control panel.

You can also install a switch between pins 6 and 7 on the 15-pin subminiature D connector to enable Micro-Ion gauge degas. To initiate the degas, close the switch. If the switch remains closed, the degas cycle will run for two minutes. You may terminate the degas cycle by opening the switch.

Pressure must be lower than 5×10^{-5} Torr (6.66×10^{-5} mbar, 6.66×10^{-3} pascal) for the degas to initiate. You cannot initiate more than two degas cycles in succession.

For RS-485 degas commands, see page 50.

Figure 3-7 Degas Button and LED on Module Control Panel

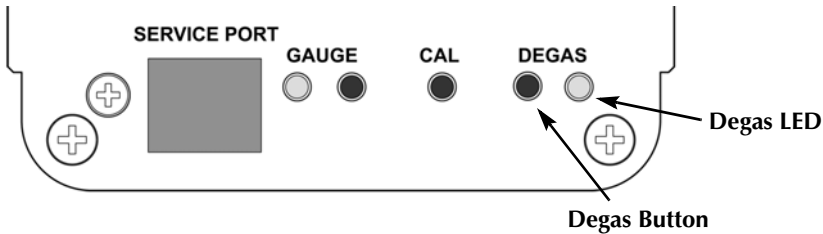


Table 3-5 Micro-Ion Gauge Status LED

Degas Status	LED State
Degas in progress	Amber

3.9 Calibrate the Module at Atmosphere

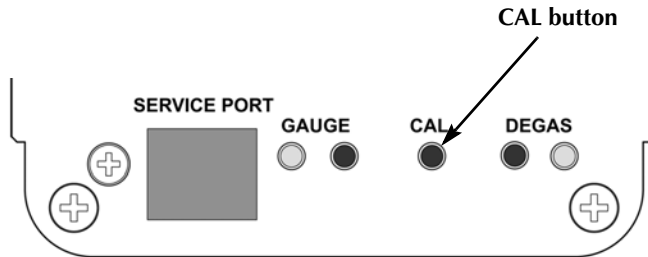
An atmospheric pressure calibration of the Conduction sensor is performed with air at the factory prior to shipment. The factory calibration sets the atmospheric calibration point ("At") to approximately 635 Torr (846 mbar, 8.46×10^4 pascal).

To ensure proper performance, the atmospheric pressure calibration of the Conduction sensor should be reset, using the process gas, once the module has been installed and is still at atmospheric pressure. Periodic resets of the atmospheric pressure calibration point will improve the accuracy and repeatability of the Conduction sensor near atmospheric pressure.

Because performance varies depending on the process gas, reset the atmospheric calibration point if a gas other than N_2 or air is being used.

Figure 3-8 Illustrates the CAL button on the module control panel.

1. Shut OFF the pump and allow pressure in the chamber to increase to atmospheric pressure. (Atmospheric pressure must be > 400 Torr [533 mbar, 5.33 kPa] to enable calibration at atmosphere.)
2. To perform the atmospheric pressure calibration, press the CAL button, or toggle pin 9 to 5 (ground) on the 15-pin subminiature D connector, or use the RS-485 interface to send the calibration (TS) command.

Figure 3-8 ATM (or CAL) Button on Module Control Panel

3.10 Calibrate the Module at Vacuum

The Conduccion sensor turns ON the Micro-Ion gauge when pressure decreases to 2.0×10^{-2} Torr (2.66×10^{-2} mbar, 2.66 pascal). If the gauge assembly has been replaced or the electronics module swapped for another module, you might need to recalibrate the Conduccion sensor at vacuum so the Micro-Ion gauge will turn ON. If the Conduccion sensor needs to be recalibrated, the optional display or other pressure output remains at a reading of > 20 mTorr (2.66×10^{-2} mbar, 2.66 pascal) when actual pressure is lower. The Conduccion sensor offset is automatically calibrated when the Micro-Ion gauge turns ON, and the pressure is sufficiently low.

Follow this procedure to recalibrate the Conduccion sensor for vacuum pressure:

Maximum calibration pressure is 1 mTorr (1.33×10^{-3} mbar, 0.13 pascal).

1. Turn ON the pump and allow the vacuum chamber to decrease to a pressure of $< 1 \times 10^{-4}$ Torr (1.33×10^{-4} mbar, 1.33×10^{-2} pascal) or a lower pressure.
2. Press the CAL button, or toggle pin 9 to 5 (ground) on the 15-pin subminiature D connector, or use the RS-485 interface to send the calibration (TZ) command.
3. The Micro-Ion Gauge should turn ON after performing the calibration.

3.11 Process Control Relay Trip Points

The module includes two relays. Each relay has a programmable trip point. When the module is shipped from the factory, trip point relays are set to a pressure of 0.0. The relays will not operate until they have been adjusted for the application.

The trip point may be set from 1×10^{-9} to 100. The pressure measurement unit restricts the trip points at the extremes of the measurement range. A built-in default hysteresis of 10% prevents oscillation around the trip point.

In the default mode, relays activate with decreasing pressure and deactivate at a 10% higher pressure than the activation pressure, as illustrated in Figure 3-9.

For RS-485 control of the trip points, see page 53.

You can reverse relay polarity, so relays activate with increasing pressure and deactivate at a lower pressure than the activation pressure, as illustrated in Figure 3-10.

Figure 3-9 Relay Behavior with Decreasing Pressure

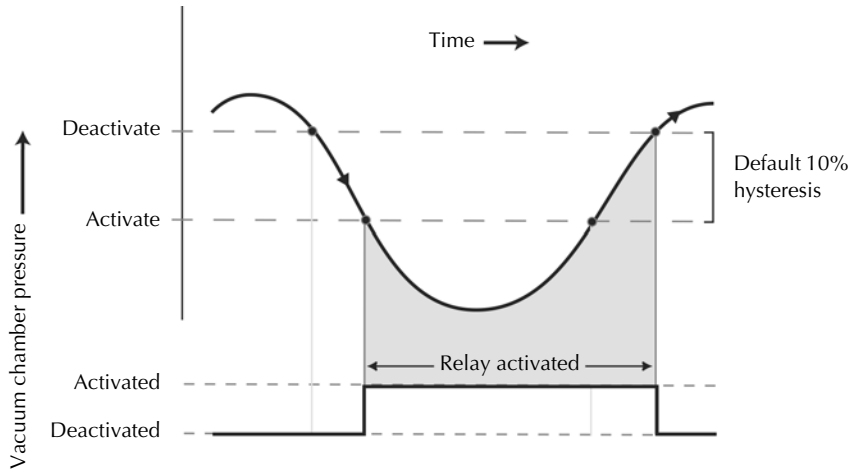
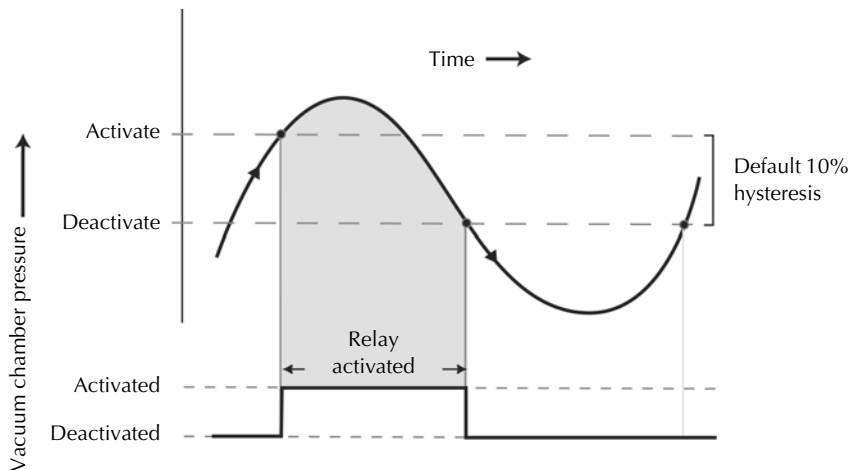


Figure 3-10 Relay Behavior with Increasing Pressure



Relay Configuration using
Trip Point Buttons and
LEDs

- 1. Make sure the module is properly installed and power is ON.
- 2. Use the optional display LEDs and recessed interface buttons to configure trip points. Use the trip points select button to activate the configuration mode and select the desired trip point for configuration.

Figure 3-11 illustrates the trip point buttons and LEDs. The corresponding LED flashes to indicate the selected trip point.

Figure 3-11 Trip Point Buttons and LEDs

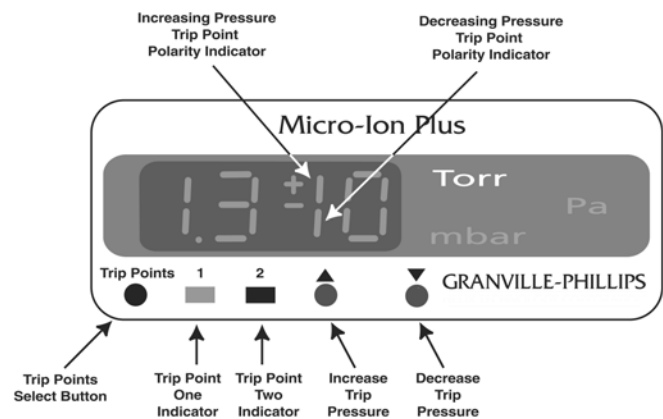


Table 3-6 indicates the status of trip point indicators during operation and configuration of trip point 1 or trip point 2.

Table 3-6 Trip Point Indicator Status

Trip Point 1 Indicator	
State	Trip Point 1 Indicator
Trip point relay 1 activated	Solid green
Trip point relay 1 deactivated	OFF
Trip point 1 configuration mode	Flashing green
Trip Point 2 Indicator	
State	Trip Point 2 Indicator
Trip point relay 2 activated	Solid green
Trip point relay 2 deactivated	OFF
Trip point 2 configuration mode	Flashing green

3. *Activate the trip point configuration mode and select the desired trip point relay.* Press the Trip Points button.
 - The first press of the button activates the configuration mode and selects trip point one (LED 1 flashes).
 - Pushing the button a second time selects trip point two (LED 2 flashes).
4. *Set the trip point pressure value.* Press the up or down arrow until the display indicates the desired activation pressure for the trip point. Press the Trip Points button to store the indicated activation pressure and select the other relay.
5. Use Table 3-7 on to record the activation pressure and activation direction for each relay.

Table 3-7 Relay Trip Points and Activation Direction

Relay	Activation Pressure for Trip Point	Activation Direction	Hysteresis (as percentage of activation pressure)
Relay 1		☐ Activate with increasing pressure ☐ Activate with decreasing pressure	
Relay 2		☐ Activate with increasing pressure ☐ Activate with decreasing pressure	

6. *If desired, select the appropriate polarity for the trip point.* If you wish to change the trip point polarity, press the up arrow or down arrow to scroll the display through the "0.0+0" display value. The upper segment of the exponent "1" digit flashes for increasing pressure, or the lower segment flashes for decreasing pressure.
7. *Return the display to the measured vacuum pressure reading.* Push the Trip Points button a third time to set the configuration values, deactivate the trip point configuration mode, and return the display to the measured vacuum pressure reading.

3.12 RS-485 Operation and Commands

This following sections of the Operation chapter provide detailed information to use the Series 356 Micro-Ion Plus Module with the RS-485 digital interface. Topics covered are:

- RS-485 Physical Layer (Section 3.13)
- RS-485 Initial Setup (Section 3.14)
- RS-485 Data Timing and Response (Section 3.15)
- RS-485 Command Structure (Section 3.16)
- RS-485 Command Sets (Section 3.17)
- RS-485 Commands (Section 3.18 and Section 3.19)

3.13 RS-485 Physical Layer**Table 3-8 RS-485 Physical Layer Specifications**

Function	Description
Arrangement	2-wire half duplex
Address range	0 to 63
Default address	01
Method for setting module address	• Use address switches to set value of 0 to 15 for address • Send SA command to set value of 10, 20, or 30 for address offset
Maximum cable length	• 4000 feet (1610 meters) • A common ground wire should connect all network devices for long cable runs
Maximum number of devices in network	31 devices
Default baud rate	19200 baud (19.2 kbaud)
Data bits	8 data bits
Stop bits and parity	1 stop bit, no parity

3.14 RS-485 Initial Setup

SA – Set the Address Offset

The default address for the module is 01. You *may* assign any address from 0 to 63 (decimal equivalent).

The address consists of the hexadecimal switch setting plus the hexadecimal SA (set address offset) value. For example, to set a value of 60 for the address, set the switch to C_{hex} (12), then send an SA value of 30_{hex} (48).

To set the address, follow these steps:

1. Send an SA (set address offset) command to the module. The example SA command sets the address offset to 30_{hex} (48):

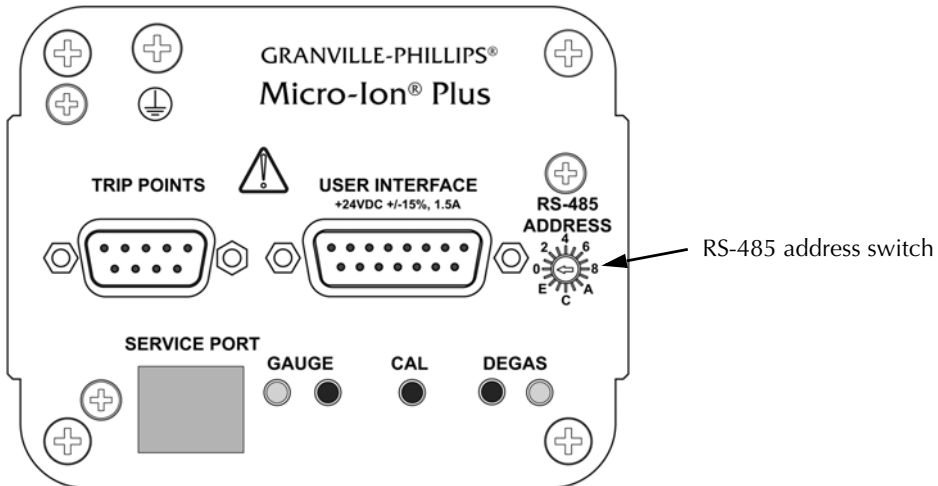
Example SA command from host: #01SA30↵

Response from module: *01 PROGM OK↵

Valid SA command values are 00, 10_{hex} (16), 20_{hex} (32), or 30_{hex} (48).

2. Use the rotary switch located on the top of the module to set a value of 0 to 15 (see Figure 3-12).
 - Set the switch on one of the unlabeled marks for the values 1, 3, 5, 7, 9, B_{hex} (11), D_{hex} (13), or F_{hex} (15).
 - Set the switch on one of the labeled marks for the values 0, 2, 4, 6, 8, A_{hex} (10), C_{hex} (12), or E_{hex} (14).
3. Cycle power to the module to activate the new address offset.

Figure 3-12 Address Switch



SB – Set the Baud Rate

The example set baud rate (SB) command sequence sets the baud rate to 2400 baud:

Example SB command from host: `#01SB2400␣`

Response from module: `*01 PROGM OK␣`

Allowable SB values are 1200, 2400, 4800, 9600, 19200 (default), or 38400 baud.

SP – Set the Parity

The example set parity (SP) command sequence sets parity at seven data bits, odd parity:

Example SP command from host: `#01SPO␣`

Response from module: `*01 PROGM OK␣`

- An SPO command sets parity at seven data bits, odd parity.
- An SPE command sets parity at seven data bits, even parity.
- An SPN command sets parity at eight data bits, no parity. This is the default parity.

3.15 Data Timing and Response

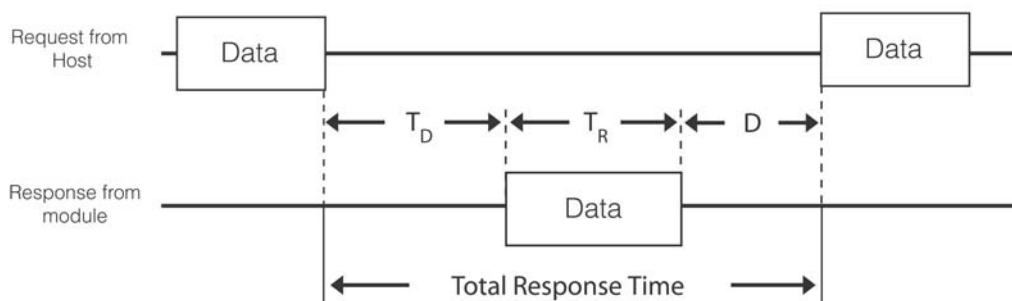
The module communicates using half-duplex mode. The host issues a command then waits for a response from the module.

Figure 3-13 illustrates the request and response data timing sequence, including:

- The request sent by the host / response sent by the module.
- Minimum and maximum time duration from the end of the receipt of the request to the start of the response (T_D).
- Time required for the module to process and respond with the data (T_R).
- Time for the module to switch between transmit and receive modes (D).

Table 3-9 lists data timing and response delay limits. The time required for the module to process and send the response depends on the baud rate and command type, as listed in Table 3-10 and Table 3-11.

The minimum response time of the module to a request is 1.2 msec. The host must switch from transmit to receive mode in 12.0 msec or less to ensure proper receipt of response data packets from the module. The host must wait a minimum of 200 μ sec after receiving the response before sending a new request command.

Figure 3-13 Data Timing and Response Delays**Table 3-9 Data Timing and Response Delay Limits**

Timing Segment	Time Limit
Time T_D (time for host to switch from transmit to receive)	1.2 msec minimum 12.0 msec maximum
Time T_R (data processing and response time)	$\left(\frac{1}{\text{Baud}} \times 130\right) + \text{Time}_P$
Time T_P (data processing time)	See Table 3-11
Time D (time for module to switch from transmit to receive)	200 μsec minimum 1.0 msec maximum
Total response time	Time $T_D + \text{Time}_R + \text{Time } D$

Table 3-10 Baud Rate and Data Response Time

Baud Rate	Response Time
1200 baud	108 msec
2400 baud	54 msec
4800 baud	27 msec
9600 baud	13.5 msec
19200 baud (default)	6.75 msec
38400 baud	3.3 msec

Table 3-11 Task and Data Processing Time

Task	Data Processing Time (T _p)
Read commands	0 sec
Write commands	5 msec
FAC (reset values to factory defaults)	28.0 msec
RST (reset module to power-up state)	No response

The response time from the module depends on the baud rate and the task that the host commands the module to perform.

Table 3-10 on page 43 lists baud rates and corresponding response times.

The module driver automatically shuts OFF 80 μ sec after returning its response to the host.

3.16 RS-485 Command Structure

RS-485 commands require entry of integer values, hex code values (such as "0F"), values in engineering notation (such as "2.00E+02"), and ASCII character strings.

Table 3-12 explains the RS-485 command structure. The command should not include a line feed with the carriage return. Including a line feed adds an extra character and may cause a garbled response from the module.

The $\downarrow$ symbol at the end of the command represents the carriage return (CR), which is entered as hex code 0D or, if you're using a terminal, by simultaneously pressing the "Control" and "M" keys.

Table 3-12 RS-485 Command Structure

Address Field	Command Field	Data Field	Carriage Return
#XX	Character string for command from host	Character string data required to execute command	$\downarrow$
"XX" is 2-digit address	First character is: • PC = Process control • S = Set • T = Calibrate • R = Read or reset • I = Ion gauge • V = Version • K = Keyboard	Data may include: • Hex code • Pressure value in engineering notation • ASCII character string	• Enter hex code "0D" • Simultaneously press "Control" and "M" keys

3.17 RS-485 Command Sets

Table 3-13 lists RS-485 commands that provide pressure values or other information without affecting module operation.

Table 3-14 lists RS-485 commands that may affect module operation and have default values.

Table 3-13 RS-485 Command Set for Values not Affecting Module Operation

Command	Command Type	Non-volatile	Change After Reset	Data Returned	Can be Locked	See Page #
IGS	Read	No	No	Micro-Ion gauge ON/OFF status	No	47
KBS	Read	Yes	No	Keyboard lock/unlock status	No	49
RD	Read	No	No	Vacuum pressure	No	47
RE	Read	No	No	Micro-Ion gauge emission current	No	47
RF	Read	No	No	Micro-Ion gauge filament status	No	47
RS	Read	No	No	Module status RS-485 string	No	48
VER	Read	Yes	No	Firmware version	No	48

Table 3-14 RS-485 Command Set for Commands Affecting Module Operation

Command	Command Type	Non-volatile	Default	Change After Reset	Data Returned	Can be Locked	See Page #
DG	Read/write	No	OFF (0)	No	Confirm or state	No	50
FAC	Write	Yes	Factory defaults	Yes	Confirm	No	56
IDT	Read/write	Yes	0 seconds	No	Confirm or state	No	51
IG	Read/write	No	ON (1)	No	Confirm or state	No	50
IGM	Read/write	Yes	ON (1)	No	Confirm or state	No	50
KBL	Write	Yes	Keyboard is unlocked	No	Confirm	No	55
KBU	Write	Yes	Keyboard is unlocked	Yes	Confirm	No	56
PC	Read/write	Yes	0.0	No	Confirm or state	No	53
PCH	Read/write	Yes	Relay deactivates at 10% above activation pressure	No	Confirm or state	No	53
PCP	Read/write	Yes	Relay activates with decreasing pressure	No	Confirm or state	No	53
RST	Write	No	None (not applicable)	No	None	No	54
SA	Write	Yes	00	Yes	Confirm	Yes	41
SB	Write	Yes	19200 baud	Yes	Confirm	Yes	42
SER	Read/write	Yes	1.0 x 10 ⁻⁵ Torr 1.33 x 10 ⁻⁵ mbar 1.33 x 10 ⁻³ pascal	No	Confirm or state	Yes	52
SF	Read/write	Yes	- Manual (MAN) mode for tungsten filaments - Alternating (ALT) mode for yttria-coated iridium filaments	No	Confirm or state	Yes	52
SP	Write	Yes	8 data bits No parity	Yes	Confirm	Yes	42
SU	Write	Yes	Torr, mbar, or pascal as specified by customer	Yes	Confirm	Yes	54
TLU	Write	Yes	Functions unlocked	No	Confirm	No	54
TS	Write	Yes	Factory setting	No	Confirm	No	54
TZ	Write	Yes	Automatically set	No	Confirm	No	54
UNL	Write	Yes	Functions unlocked	No	Confirm	No	55

3.18 RS-485 Commands - Set 1

The following seven RS-485 commands provide pressure values or other information which do not affect the operation of the Micro-Ion Module. See Table 3-13 on page 45.

RD - Read Vacuum Pressure

The example read pressure (RD) command causes the module to return a value that indicates vacuum pressure is 1.50×10^{-2} :

Example RD command from host: `#01RD␣`

Example response from module: `*01 1.50E-02␣`

- The returned value is in the unit of measure that has been set for the module.
- If the returned value is not a valid representation of pressure, see *RS – Read the Module Status RS-485 Strings* on page 48.
- You may also use the optional numeric display to read setpoint pressure and error codes.

IGS - Read the Micro-Ion Gauge ON/OFF Status

When you send the ion gauge status (IGS) command, the module returns a character string that indicates the ON/OFF status of the Micro-Ion gauge.

IGS command from host: `#01IGS␣`

Example response from module: `*01 1 IG ON␣`

- The “1 IG ON” response means the Micro-Ion gauge is ON.
- The “0 IG OFF” response means the Micro-Ion gauge is OFF.

RF - Read Micro-Ion Gauge Filament Status

The read filament (RF) RS-485 command causes the module to return a character string indicating the operation mode of the gauge filament.

RF command from host: `#01RF␣`

Example response from module: `*01 FIL SF1␣`

- The “FIL SF1” response means filament 1 is operating.
- The “FIL SF2” response means filament 2 is operating.
- The “FILBOTH” response means a degas cycle is in progress.

RE - Read Micro-Ion Gauge Emission Current

Send the read emission (RE) command to read the Micro-Ion gauge emission current. Table 3-15 lists module responses to the RE command.

RE command from host: `#01RE␣`

Example response from module: `*01 4.0MA EM␣`

Table 3-15 Responses to RE Command

Response	Description
4.0MA EM	Micro-Ion gauge is in high-emission mode (4.0 mA current)
0.1MA EM	Micro-Ion gauge is in low-emission mode (0.1 mA current)
0.02MA EM	Micro-Ion gauge is in low-emission mode 0.02 mA current)
15MA EM	Micro-Ion gauge degas cycle is in progress (see <i>Degas the Micro-Ion Gauge</i> on page 34)
0 IG OFF	Micro-Ion gauge is OFF (see <i>Turn the Micro-Ion Gauge ON or OFF</i> on page 31 and <i>IGS - Read the Micro-Ion Gauge ON/OFF Status</i> on page 47)

VER – Read the Firmware Version

The example read firmware version (VER) command causes the module to return a value that represents the Granville-Phillips internal part number 14851 and firmware revision 07 for the module:

Example VER command from host: #01VER↵

Example response from module: *01 14851-07↵

- The first five digits (preceding the dash) are the internal part number. This number will change depending on the module.
- The last two digits (following the dash) are the firmware version.

RS – Read the Module Status RS-485 Strings

Send the read status (RS) command to read RS-485 character strings that indicate the module operating status.

RS command from host: #01RS↵

Example response from module: *01 00 ST OK↵

The example response indicates the module is operating normally.

Table 3-16 lists module responses to the RS command.

Table 3-16 Module Status RS-485 Strings

Response	Cause	Solution
00 ST OK	Module is operating normally.	No solution necessary.
01 CGBAD	Conductron sensor is inoperable.	• Cycle power to the module. • If necessary, replace the gauge assembly (see page 66).
05 IG HV	Micro-Ion gauge grid voltage failure.	Cycle power to the module or send IG1 command to clear the status response (see page 61).
06 IG EM	Micro-Ion gauge emission failure.	
07 IGFIL	One Micro-Ion gauge filament is open.	• If SF command is set to AUTO, ALT, or Y, operation automatically switches to the other filament (see page 52). • If SF command is set to MAN or T, send IG0 command, then send IG1 command to switch filaments (see page 50).
08 POWER	Power cycle has occurred, module is starting up.	No solution necessary. The "00 ST OK" string will be generated next if the module is operating properly.

KBS – Read Keyboard Lock/Unlock Status

Use the keyboard status (KBS) command to determine whether the CAL (or ATM), Gauge, Degas, and Trip Points buttons are locked or unlocked.

KBS command from host: #01KBS↵

Response from module: *01 1 KB OFF↵

- The "1 KB OFF" response means the ATM (or CAL), Gauge, Degas, and Trip Points buttons are unlocked.
- The "1 KB ON" response means the ATM (or CAL), Gauge, Degas, and Trip Points buttons are locked.

3.19 RS-485 Commands - Set 2

The following eighteen RS-485 commands may affect the operation of the Micro-Ion Module, and have default values. See Table 3-14 on page 46.

DG - Degas Commands

The RS-485 command includes the alpha characters "D" and "G" and the numeric value 0 (terminate) or 1 (initiate).

To degas the Micro-Ion gauge, follow these steps:

1. Turn ON the Micro-Ion gauge. (See *Turn the Micro-Ion Gauge ON or OFF* on page 31.)
2. Make sure vacuum pressure is lower than 5×10^{-5} Torr (6.66×10^{-5} mbar, 6.66×10^{-3} pascal).
3. Send the degas gauge initiate (DG1) command to initiate the Micro-Ion gauge degas cycle. The time for gauge degas is two minutes.

DG1 command from host: #01DG1↵

Response from module: *01 PROGM OK↵

4. If you want to terminate the Micro-Ion gauge degas cycle before it is completed, send the degas gauge terminate (DG0) command.

DG0 command from host: #01DG0↵

Response from module: *01 PROGM OK↵

IG - Ion Gauge Commands

The RS-485 command includes the alpha characters "I" and "G" and the numeric value 0 (OFF) or 1 (ON).

Send the ion gauge OFF (IG0) command to turn OFF the Micro-Ion gauge.

IG0 command from host: #01IG0↵

Response from module: *01 PROGM OK↵

If you send the IG0 command, the Micro-Ion gauge remains OFF until you send an IG1 command to turn it ON.

Send the ion gauge on (IG1) command to turn the Micro-Ion gauge ON.

IG1 command from host: #01IG1↵

Response from module: *01 PROGM OK↵

IGM - Ion Gauge Mode

If the Micro-Ion gauge is OFF, the IGM (ion gauge mode) command can be used to enable or disable pressure indications from the Conductron sensor.

The IGM RS-485 command includes the alpha characters "I", "G", and "M" and the alphanumeric 0 (OFF) or 1 (ON).

Example IGM command from host: #01IGM1↵

Response from module: *01 PROGM OK↵

- The 0 switch disables pressure indications from the Conductron sensor while the Micro-Ion gauge is OFF. The display reads OFF.
- The 1 switch enables pressure indications from the Conductron sensor while the Micro-Ion gauge is OFF. (Minimum output is 1×10^{-3})

Use the “S” switch to read the pressure output status.

IGMS command from host: #01IGMS↵

Example response from module: *01 0 ALL↵

- The “0 ALL” character string means all pressure indications are *disabled* when the Micro-Ion gauge is OFF.
- The “1 IG” character string means pressure indications are *enabled* when the Micro-Ion gauge is OFF.

IDT - Set or Read the Micro-Ion Gauge Delay Time

The ion gauge delay time (IDT) command changes the amount of time the Micro-Ion gauge waits to turn ON after pressure has decreased to a point where the gauge can operate. This is useful to allow sufficient time for corrosive gases to be pumped out of the chamber before the Micro-Ion gauge hot filament is turned ON.

The example IDT command sets the additional delay time to 20 seconds.

Example IDT command from host: #01IDT 20↵

Response from module: *01 PROGM OK↵

- The command includes the alpha characters “I”, “D”, and “T” and the numeric value representing the number of seconds.
- Valid values are 0 to 600 seconds.
- The Micro-Ion gauge will wait the number of seconds specified in the IDT command, *plus* the software-defined delay, before turning on with decreasing pressure.
- With increasing pressure, the Micro-Ion gauge will turn OFF when pressure increases to 30 mTorr (3.99×10^{-2} mbar, 3.99 pascal).

If you send the IDT command without the time value, the module returns a character string that represents the Micro-Ion gauge delay time.

IDT command from host: #01IDT↵

Example response from module: *01 60 IDT↵

The example response means the delay time is set to 60 seconds. If the SF command is set to T (tungsten), the response does *not* include the additional software-determined delay.

During the delay, outputs indicate pressure as measured by the Conductron sensor.

SF - Set Filament

The Set Filament (SF) RS-485 command can be used to change the filament operation mode of the Module. The best filament operational mode for the type of Micro-Ion Gauge in the Module you are using was set at the factory. The following RS-485 command (SF) example sets the gauge filament to manual mode.

Example SF command from host: #01SFMAN↵

Response from module: *01 PROGM OK↵

Table 3-17 lists other modes that can be set.

Table 3-17 Filament Mode Settings

Setting	Description
SFALT	- Sets the Micro-Ion gauge to alternating mode (see Table A-2 on page 67) - Default mode for Micro-Ion gauge with yttria-coated iridium filaments
SFAUTO	Sets the Micro-Ion gauge to automatic mode (see Table A-2 on page 67)
SFMAN	- Sets the Micro-Ion gauge to manual mode (see Table A-3 on page 67) - Default mode for Micro-Ion gauge with tungsten filaments
SFT	- Sets the Micro-Ion gauge to manual mode (see Table A-3 on page 67) - Sets IDT command to a value of 10 seconds (see page 51)
SFY	- Sets the Micro-Ion gauge to alternating mode (see Table A-2 on page 67) - Sets IDT command to a value of 0 seconds (see page 51)

SER - Set Emission Range

Use the RS-485 set emission range (SER) command to set or read the pressure at which the Micro-Ion gauge switches from high to low emission current with increasing pressure. The emission switching pressure may need to be changed to adapt to processing pressure.

Example SER command from host:#01SER 1.00E-06↵

Response from module: *01 PROGM OK↵

The example SER command sets a value of 1×10^{-6} for the pressure at which the Micro-Ion gauge switches from high to low emission current.

To read the emission current switch point, send an SER command without the pressure value.

Example SER command from host:#01SER↵

Example response from module: *01 1.00E-05↵

PC - Process Control

1. Make sure the module is properly installed and the power is ON.
2. Use Table 3-7 to record the activation pressure, activation direction, and hysteresis for each relay.
3. Refer to the following example process control (PC) command sequence to program the pressure at which the relay will activate:

The example PC command sequence causes relay 1 to activate when vacuum pressure decreases to 1.01×10^{-1} .

Example PC command from host: #01PC1 1.01E-01↵

Response from module: *01 PROGM OK↵

The "PC1" command identifies process control relay 1. The "PC2" command identifies process control relay 2.

PCP - Process Control Polarity

In default mode, relays activate with decreasing pressure and deactivate at a 10% higher pressure than the activation pressure, as illustrated in Figure 3-9 on page 37.

The example process control polarity (PCP) command causes relay 1 to activate with increasing pressure.

Example PCP command from host: #01PCP1 +↵

Response from module: *01 PROGM OK↵

- The "PCP1" command identifies process control relay 1. The "PCP2" command identifies process control relay 2.
- The + sign causes the relay to activate with increasing pressure.
- The – sign causes the relay to activate with decreasing pressure.

PCH - Process Control Hysteresis

- Default hysteresis is 10% below activation pressure if the relay activates with increasing pressure (+).
- Default hysteresis is 10% above activation pressure if the relay activates with decreasing pressure (–).

The example process control hysteresis (PCH) command causes relay 1 to deactivate when increasing pressure is 20% lower than the activation pressure, if relay 1 polarity activates with increasing pressure (+).

Example PCH command from host: #01PCH1 20↵

Response from module: *01 PROGM OK↵

- The "PCH1" command identifies process control relay 1. The "PCH2" command identifies relay 2.

Valid values are any percentage, from 5% to 100%, that is divisible by 5.

**TS - Calibrate the Module
at Atmosphere**

Example TS command from host: #01TS↵

Example response from module: *01 PROGM OK↵

If the module returns a "RANGE ER" message in response to the TS command, the atmospheric calibration has failed. The raw sensor voltage is outside the normal range for the Conductron sensor. See Table 3-20 on page 57 to troubleshoot the problem.

If the module has an optional display, the display will indicate "-- --" during the push-button calibration.

**TZ - Calibrate the Module
at Vacuum**

Example TZ command from host: #01TZ↵

Example response from module: *01 PROGM OK↵

If the module returns a "RANGE ER" message in response to the TZ command, the vacuum pressure calibration has failed. The raw sensor voltage is outside the normal range for the offset of the Conductron sensor. See page 57 to troubleshoot the problem.

The Micro-Ion gauge will turn ON after the TZ command has been sent.

SU – Set the Pressure Unit

The example set unit (SU) command sets the pressure unit to Torr:

Example SU command from host: #01SUT↵

Response from module: *01 PROGM OK↵

- An SUT command sets the pressure unit to Torr.
- An SUM command sets the pressure unit to millibar.
- An SUP command sets the pressure unit to Pascal.

**RST – Reset the Module
to Power-up State**

The reset (RST) command resets the module to power-up status.

RST command from host: #01RST↵

Response from module: None

Sending the RST command has the same effect as cycling power to the module. Communication is re-enabled two seconds after you've sent the RST command.

**TLU – Toggle the Locked
Functions**

In the default operating mode, all interface functions are unlocked. Use the toggle lock/unlock (TLU) command to lock or unlock any of the interface functions listed in Table 3-18.

Table 3-18 Interface Functions Affected by TLU Command

Command	Interface Function	Instructions:
SER	Set emission current threshold for Micro-Ion gauge	Page 52
SA	Set address offset	Page 41
SB	Set baud rate	Page 42
SP	Set parity	Page 42
SU	Set pressure unit	Page 54
KBL	Lock control functions (keyboard)	Page 55
KBU	Unlock control functions (keyboard)	Page 56
FAC	Reset values to factory defaults	Page 56

The module processes the command, then returns a character string that indicates whether or not interface functions are unlocked.

TLU command from host: #01TLU↵
 Response from module: *01 1 UL ON↵

- The “1 UL ON” response means interface functions are locked.
- The “0 UL OFF” response means interface functions are unlocked.

UNL – Unlock the Interface Functions

If you’ve used the TLU command to lock interface functions listed in Table 3-18, you must send the unlock interface functions (UNL) command to unlock a function before reprogramming it.

UNL command from host: #01UNL↵
 Response from module: *01 PROGM OK↵

- If you send the UNL command while interface functions are already unlocked, the module returns a “SYNTAX ER” response (see page 57).
- You can unlock all locked functions by sending the TLU command.
- If you attempt to reprogram a locked function without sending the UNL or TLU command, the module returns a “LOCKED” response (see page 57).

KBL – Lock the Keyboard

The keyboard lock (KBL) command enables you to lock the CAL (or ATM), Gauge, Degas, and Trip Points buttons on the module.

KBL command from host: #01KBL↵
 Response from module: *01 PROGM OK↵

After you’ve locked the buttons, you must use the KBU command to unlock them.

You may also ground pin 11 on the 15-pin subminiature D connector to lock the CAL, Gauge, Degas, and Trip Points buttons on the module.

To unlock the buttons, remove the pin 11 ground.

If you've locked the buttons by grounding pin 11, you cannot use the KBU command via RS-485 to unlock them. Remove the ground before sending the KBU command.

KBU – Unlock the Keyboard

After you've locked the CAL (or ATM), Gauge, Degas, and Trip Points buttons, use the keyboard unlock (KBU) command to unlock them.

KBU command from host: #01KBU↵

Response from module: *01 PROGM OK↵

If you've locked the buttons by grounding pin 11 on the 15-pin subminiature D connector, the KBU command will not change the locked state. Remove the ground before sending the KBU command.

FAC – Reset the Values to Factory Defaults

Table 3-19 lists default settings for the RS-485 version of the module. After you've reconfigured the module, you must cycle power to the module before the new settings will take effect.

You may restore parameters listed in Table 3-19 to their default values by sending a factory reset (FAC) command.

FAC command from host: #01FAC↵

Response from module: *01 PROGM OK↵

Reset command from host: #01RST↵

Response from module: None

Table 3-19 Factory Default Settings for RS-485 Module

Parameter	Default Setting
Digital communication	- Parity: 8 data bits, no parity, 1 stop bit - Baud rate: 19.2 kbaud - Module address: 01 - Address offset: 00
Emission current threshold	5×10^{-6} Torr (7×10^{-6} mbar, 7×10^{-4} pascal) with decreasing pressure 1×10^{-5} Torr (1×10^{-5} mbar, 1×10^{-3} pascal) with increasing pressure
Relays	- Trip point: Sets to 0.0 vacuum pressure; relay must be configured to be operable - Hysteresis and activation direction: Relay activates at 10% above vacuum pressure trip point (–) with decreasing pressure
Pressure output units and optional pressure display	Pressure unit is Torr, mbar, or pascal as specified on sales order and catalog number tag

3.20 RS-485 Error Responses and Display Error Codes

- Table 3-20 lists error responses that the module returns if you enter a command improperly or if a value in a command is out of range.
- Table 3-21 lists error codes produced by the optional pressure display.

Table 3-20 RS-485 Error Responses

Response*	Possible Causes	Solution
?XX RANGE ER	Pressure value in TS, TZ, SER, or IDT command is outside valid limits.	• Make sure atmospheric pressure is > 400 Torr (533 mbar, 53.3 kPa), then re-send the TS command (see page 54). • Make sure vacuum pressure is $< 1 \times 10^{-3}$ Torr (1.33×10^{-3} mbar, 0.13 pascal), then re-send the TZ command (see page 54). • Make sure the pressure represented in SER command is $> 1.0 \times 10^{-7}$ Torr (1.3×10^{-7} mbar, 1.3×10^{-5} pascal) and $< 1.0 \times 10^{-4}$ Torr (1.3×10^{-4} mbar, 1.3×10^{-2} pascal), then re-send the SER command (see page 52). • Make sure the number of seconds in the IDT command is 0 to 600, then re-send the IDT command (see page 51).
?XX SYNTAX ER	• Command was improperly entered. • Module does not recognize command syntax.	Re-enter the command using the proper character string.
?XX 9.99E+09	Module cannot indicate a valid pressure value.	• Send the RS command to determine the module status (see page 48). • If necessary, replace the gauge assembly (see page 66).
?XX LOCKED	User attempted to reprogram a locked function without sending TLU or UNL command.	Send the TLU or UNL command, then re-send the attempted command (see page 54).
?XX INVALID	Command syntax is correct, but command cannot be executed in current state (for example, user attempts to initiate degas cycle while Micro-Ion gauge is OFF).	Change the module to a valid state for command, then re-send the attempted command.

*In each response "XX" represents the module address.

Table 3-21 Display Error Codes

Response	Cause	Solution
Er #1	Conductron sensor is inoperable.	Replace the gauge assembly (see page 66).
Er #5	Micro-Ion gauge grid voltage failure.	Cycle power to the module.
Er #6	Micro-Ion gauge emission failure.	Cycle power to the module.

4.1 Customer Service

Some minor difficulties are readily corrected in the field. If the product must be returned to the factory for service, request a Return Material Authorization (RMA) from MKS, which can be completed at <https://www.mksinst.com/service/servicehome.aspx>. Do not return products without first obtaining an RMA. In most cases a hazardous materials disclosure form is required. The MKS Customer Service Representative will advise you if the hazardous materials document is required.

When returning products to MKS, be sure to package the products to prevent shipping damage. Shipping damage on returned products as a result of inadequate packaging is the Buyer's responsibility.

For Customer Service / Technical Support:

MKS Pressure and Vacuum Measurement Solutions

MKS Instruments, Inc.
6450 Dry Creek Parkway
Longmont, Colorado 80503 USA
Tel: 303-652-4400
Fax: 303-652-2844
Email: mks@mksinst.com

MKS Corporate Headquarters

MKS Instruments, Inc.
2 Tech Drive, Suite 201
Andover, MA 01810 USA
Tel: 978-645-5500
Fax: 978-557-5100
Email: mks@mksinst.com

Damage Requiring Service

Shut off power to the module and refer servicing to qualified service personnel under the following conditions:

- The product has been exposed to rain or water.
- The product does not operate normally, even if you have followed the operation instructions. Operate only those controls that are covered in the instruction manual.
- The product has been dropped or the module enclosure has been damaged.
- The product exhibits a distinct change in performance.

If the module requires repair:

- See Section 4.5 Returning a Micro-Ion Module for Service on page 66
- Phone **1-303-652-4400** or email mks@mksinst.com.

4.2 Troubleshooting

If any of the conditions described on page 59 have occurred, troubleshooting is required to determine the repairs that are necessary.

Precautions

Because the module contains static-sensitive electronic parts, follow these precautions while troubleshooting:

- Use a grounded, conductive work surface. Wear a high impedance ground strap for personal protection.
- Do not operate the module with static sensitive devices or other components removed from the product.
- Do not handle static sensitive devices more than absolutely necessary, and only when wearing a ground strap.
- Use a grounded, electrostatic discharge safe soldering iron.



WARNING

Substitution or modifying parts can result in severe product damage or personal injury due to electrical shock or fire.

- Install only those replacement parts that are specified by Granville–Phillips.
- Do not install substitute parts or perform any unauthorized modification to the module.
- Do not use the module if unauthorized modifications have been made.

Symptoms, Causes, and Solutions

Table 4-1 lists failure symptoms, causes, and solutions indicated by something other than an RS-485 error message from the module.

Table 4-1 Troubleshooting Symptoms and Causes

Symptom	Causes	Solutions
Gauge indicator does not illuminate.	- Power supply is disconnected, OFF, or inadequate for load. - A switching power supply may have shut down due to a current surge during power-up. - Power supply wiring is faulty. - Power supply connector is wired incorrectly.	- Make sure the power supply is ON. - Make sure the power supply can withstand the startup inrush current (see page 18). - Replace or repair the power supply wiring (see page 18). - Reconnect the wiring to the power supply connector (see page 18).
Process control trip point does not function properly.	- Trip point connector is wired incorrectly. - Trip points are set to invalid values.	- Reconnect the wiring to the trip point connector (see page 21). - Reconfigure the trip points (see page 38).
Micro-Ion gauge will not stay on.	- Vacuum chamber pressure $> 2 \times 10^{-2}$ Torr (2.66×10^{-2} mbar, 2.66 pascal). - Er #6: Micro-Ion gauge failure due to broken filament or contamination. - Er #5: Micro-Ion gauge failure due to damage, contamination, or power supply failure.	- Decrease the vacuum chamber pressure to $< 2 \times 10^{-2}$ Torr (2.66×10^{-2} mbar, 2.66 pascal). - If the filament is broken or mechanical damage has occurred, replace the gauge assembly (see page 66). - If contamination is present, degas the Micro-Ion gauge (see page 34).

Table 4-1 Troubleshooting Symptoms and Causes (continued)

Symptom	Causes	Solutions
Pressure reading is inaccurate.	• Mechanical damage has occurred. • Micro-Ion gauge electrode structure is contaminated by process in the vacuum chamber. • Vacuum leak in the chamber. • Conductron gauge out of calibration.	• If mechanical damage has occurred, return the module to the factory (see Section 4.5 Returning a Micro-Ion Module for Service on page 66). • If contamination is present, degas the Micro-Ion gauge (see page 34) or remove the gauge and sensor (see page 66) and bake out to 150 °C (302 °F). • If the vacuum chamber leaks, check the vacuum system. • Calibrate the Micro-Ion Module at vacuum pressure (see page 36).
Display reads “OFF”.	• Gauge button has been pressed, the Micro-Ion gauge is OFF. • Connection between pins 1 and 5 on the 15-pin connector is open.	• Turn ON the Micro-Ion gauge (see page 31). • Check the circuit and cables (see Table 2-2 on page 19).
Display reads “0.3–9” when the Micro-Ion gauge turns ON.	• Micro-Ion gauge collector is coated. • Electrometer in module is faulty. • Vacuum chamber pressure is $< 3 \times 10^{-10}$ Torr (3.99×10^{-10} mbar, 3.99×10^{-8} pascal).	• Perform the Micro-Ion gauge continuity test (see page 63). • Replace the gauge assembly (see Section 4.4 Replacing the Gauge Assembly on page 66). • Return the module to the factory (see Section 4.4 Replacing the Gauge Assembly on page 66).
Vacuum chamber pressure is < 20 mTorr (2.66×10^{-2} mbar, 2.66 Pa), but the pressure display is > 20 mTorr (2.66×10^{-2} mbar, 2.66 pascal) and the Micro-Ion gauge will not turn ON.	• Conductron sensor offset is not set correctly (out of calibration).	Calibrate the Micro-Ion Module at vacuum pressure (see page 36).

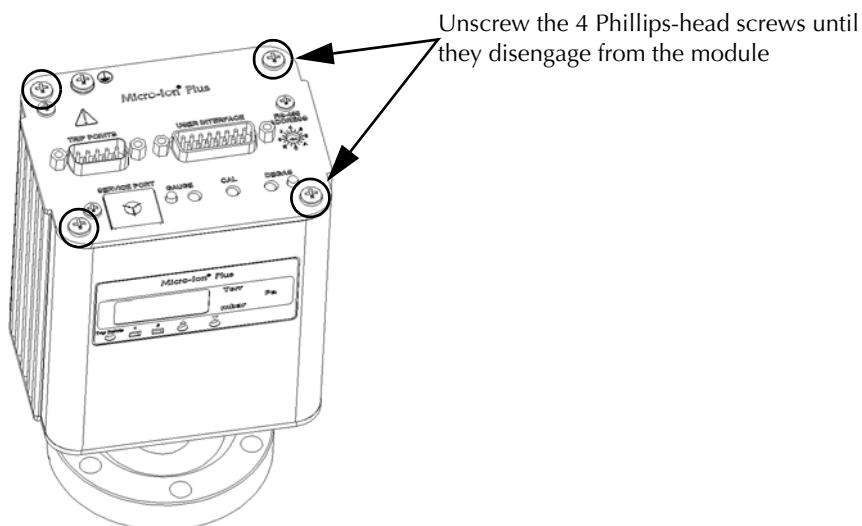
4.3 Micro-Ion Gauge Continuity Test

If a problem with pressure measurement is traced to the module, the Micro-Ion gauge can be tested with an ohmmeter. This test can detect open filaments or shorts between gauge elements, but may not detect inaccurate pressure measurement associated with vacuum leaks or other materials (contaminants) within the gauge.

The gauge can be left on the system for the test. The module electronics will be removed to gain access to the pins on the gauge.

1. Turn OFF power and disconnect all electrical connectors to the Micro-Ion Plus module.
2. Unscrew the four Phillips-head screws until they disengage from the gauge. See Figure 4-1.

Figure 4-1 Removing the Gauge, Step 2



3. *Carefully* unplug the module electronics from the gauge to expose the gauge and end plate assembly. See Figure 4-2 and Figure 4-3.

Figure 4-2 Removing the Gauge, Step 3

Carefully unplug the module electronics from the gauge

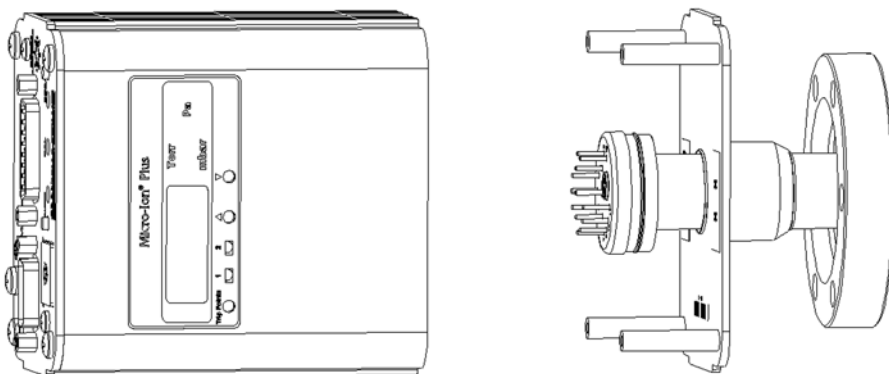


Figure 4-3 Micro-Ion Plus Replacement Gauge

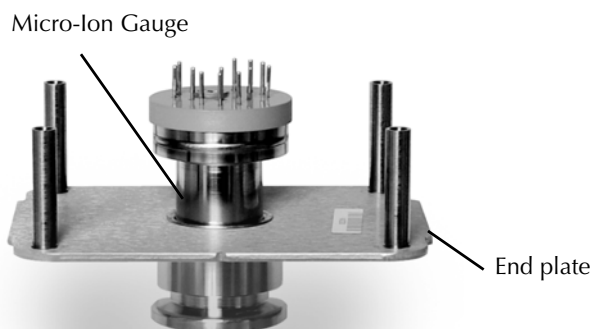
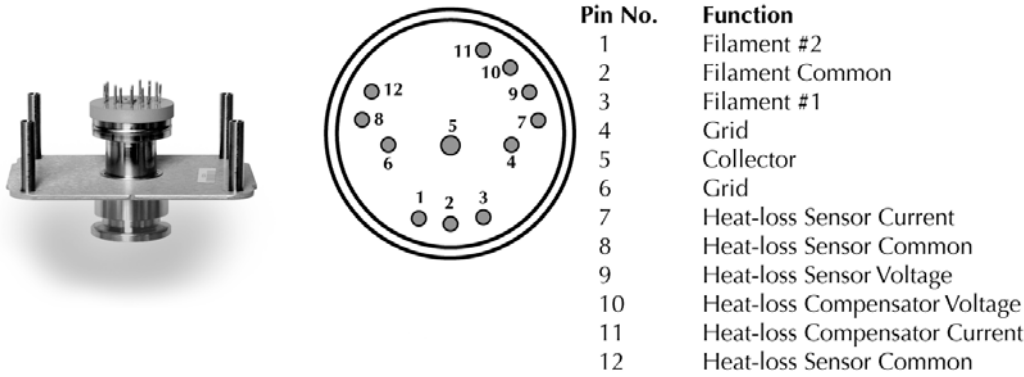


Figure 4-4 Micro-Ion Gauge Pins



4. Use a digital multimeter to measure the resistance between pins 2 and 3 (the filament pins), and between pins 1 and 2. See Figure 4-4. The readings should be $< 2 \Omega$.
5. Measure the resistance of filament pins 1, 2, or 3 to any of pins 4, 5, or 6, or the gauge case. The reading should be $> 100 M\Omega$.
6. Measure the resistance between pins 4 and 6 (the grid pins). The reading should be $< 2 \Omega$.
7. Measure the resistance of pin 5 (the collector pin) to the gauge case. The reading should be $> 100 M\Omega$.
8. Measure the resistance between pin 9 and pin 10. Resistance between heat-loss sensor wires should be $< 14 \Omega$.
9. If any of the tests result in different readings than listed above, contact MKS customer service to order a replacement gauge.

Table 4-2 Test Resistance Values

Pins	Normal Values
1 to 2	$< 2 \Omega$
2 to 3	$< 2 \Omega$
1, 2 or 3 to pins 4, 5, or 6	$> 100 M\Omega$
4 to 6	$< 2 \Omega$
pin 5 to case	$> 100 M\Omega$
9 to 10	$< 14 \Omega$

4.4 Replacing the Gauge Assembly

Do not plug in or unplug any connectors with power applied to the module. Disconnect power from the module before replacing the Micro-Ion gauge.

1. Turn OFF power and disconnect all electrical connectors to the module.
2. Remove the module from the vacuum system.
3. Unscrew the four Phillips-head screws until they disengage from the gauge. See Figure 4-1.
4. While holding the vacuum connection flange, gently unplug the gauge and end plate out of the module as shown in Figure 4-2. The gauge and end plate are shown in Figure 4-3.
5. Insert the new gauge into the module by gently inserting the gauge pins into the socket on the circuit board. Look at the gauge pin arrangement to be sure you do not have the replacement gauge and end plate assembly improperly rotated.
6. Tighten the four Phillips-head screws.
7. Install the module on the vacuum system.
8. Calibrate the Micro-Ion Module at atmosphere (see page 35).
9. If Micro-Ion gauge does not turn ON at 20 mTorr (2.66×10^{-2} mbar, 2.66 pascal) with decreasing pressure, calibrate the Micro-Ion Module at vacuum (see page 36).

4.5 Returning a Micro-Ion Module for Service

If the product must be returned to the factory for service, request a Return Material Authorization (RMA) from Granville-Phillips. Do not return products without first obtaining an RMA. In some cases a hazardous materials disclosure form may be required. The MKS/Granville-Phillips Customer Service Representative will advise you if the hazardous materials document is required.

When returning products to Granville-Phillips, be sure to package the products to prevent shipping damage. Shipping damage on returned products as a result of inadequate packaging is the Buyer's responsibility.

For Customer Service / Technical Support:

MKS Pressure and Vacuum Measurement Solutions

MKS Instruments, Inc., Granville-Phillips® Division
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Fax: 303-652-2844
Email: mks@mksinst.com

Appendix A Filament Operation

Table A-1 Micro-Ion Filament Operation Modes

Mode	Description
Automatic	- Filament logic operates filament 1 each time the Micro-Ion gauge turns ON - If filament 1 becomes inoperable, the Micro-Ion gauge switches to filament 2
Alternating (default for Micro-Ion gauge with yttria-coated iridium filaments)	- Filament logic alternates between filament 1 and filament 2 each time the Micro-Ion gauge turns ON - If a filament becomes inoperable, the module switches to the operable filament - Default mode for Micro-Ion gauge with yttria-coated iridium filaments
Manual (default for Micro-Ion gauge with tungsten filaments)	- Filament logic operates filament 1 each time the Micro-Ion gauge turns ON - If filament 1 becomes inoperable, press the gauge button to turn filament 2 ON (gauge LED will flash red) - After the gauge button has been pressed, filament 2 turns ON - Default mode for Micro-Ion gauge with tungsten filaments

Table A-2 Micro-Ion Gauge, Gauge LED, and Filament States in Automatic or Alternating Mode

Micro-Ion Gauge State	Gauge LED State	Filament State
OFF automatically	Amber	Micro-Ion gauge has automatically turned OFF
OFF and disabled	Flashing amber	Micro-Ion gauge is disabled
ON	Green	- Filament 1 or 2 is ON - Both filaments are operable
ON	Flashing green	Filament 1 or 2 is inoperable
OFF	Red	Filaments 1 and 2 are inoperable

Table A-3 Micro-Ion Gauge, Gauge LED, and Filament States in Manual Mode

Micro-Ion Gauge State	Gauge LED State	Filament State
OFF automatically	Amber	Micro-Ion gauge has automatically turned OFF
OFF and disabled	Flashing amber	Micro-Ion gauge is disabled
ON	Green	Filament 1 is ON
ON	Flashing green	Filament 2 is ON
OFF	Flashing red	- Filament 1 is inoperable - Press gauge button to switch to filament 2
OFF	Red	Filaments 1 and 2 are inoperable

Figure A-1 Automatic Filament Switching Flow Chart

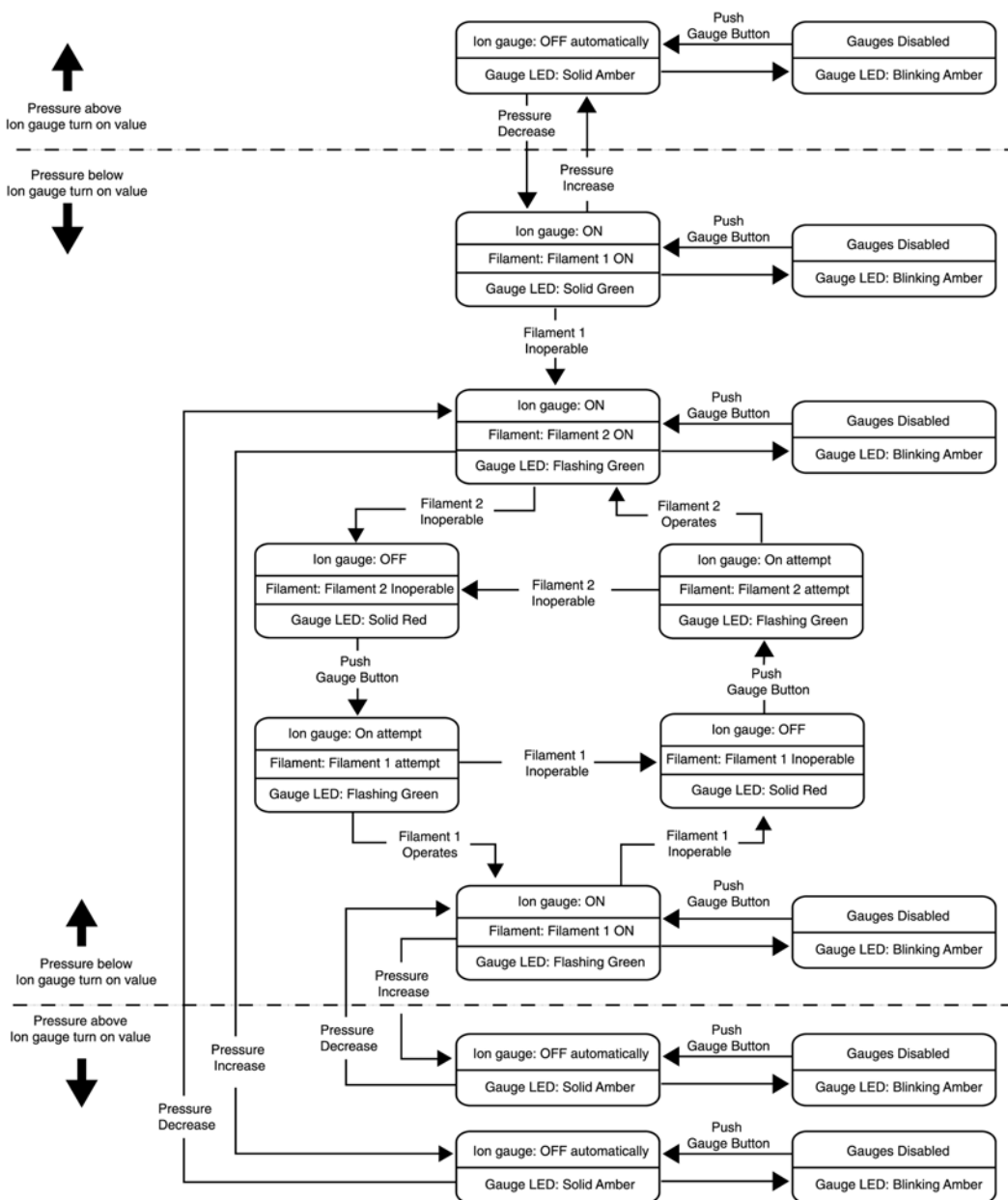


Figure A-2 Alternating Filament Switching Flow Chart

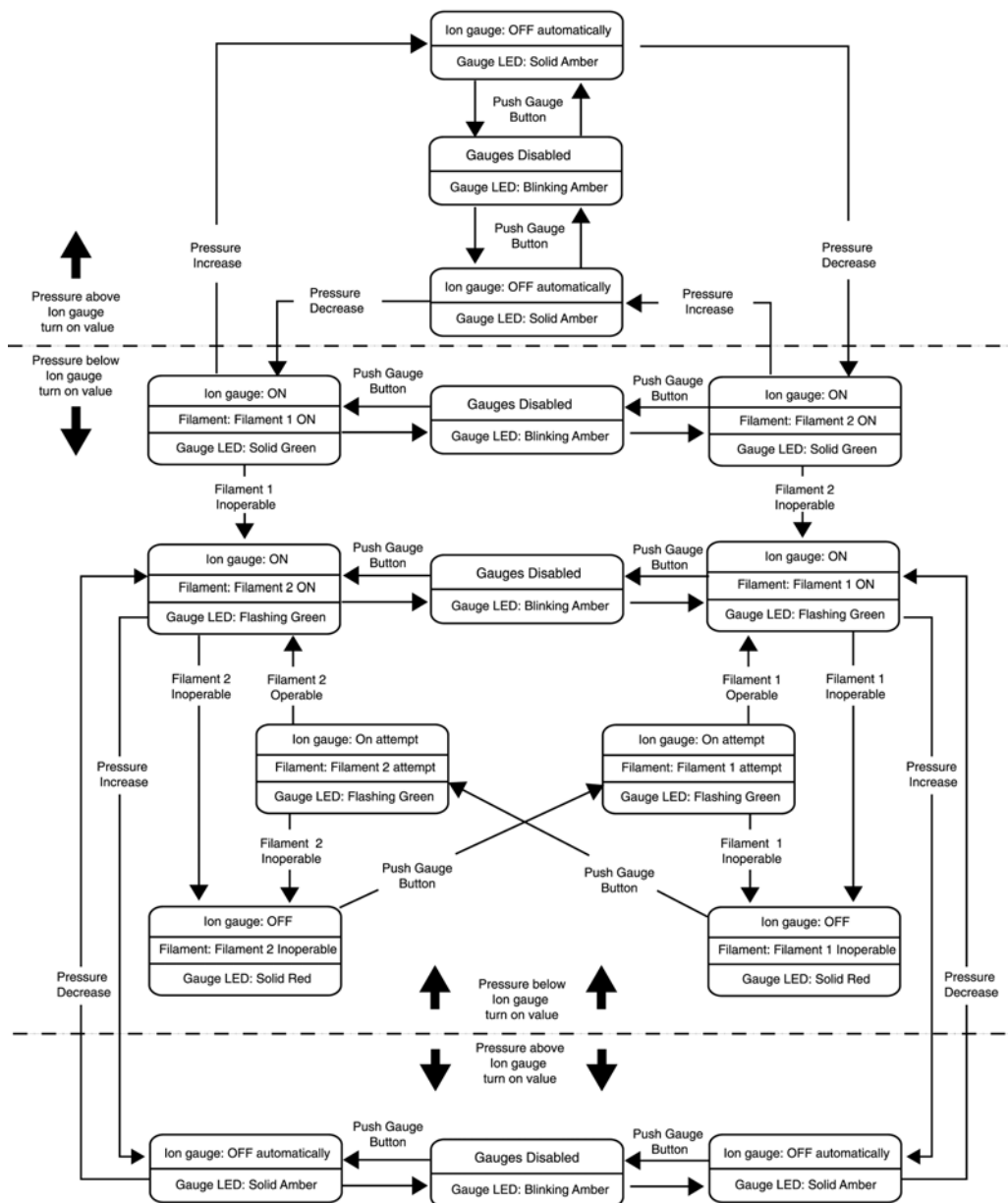
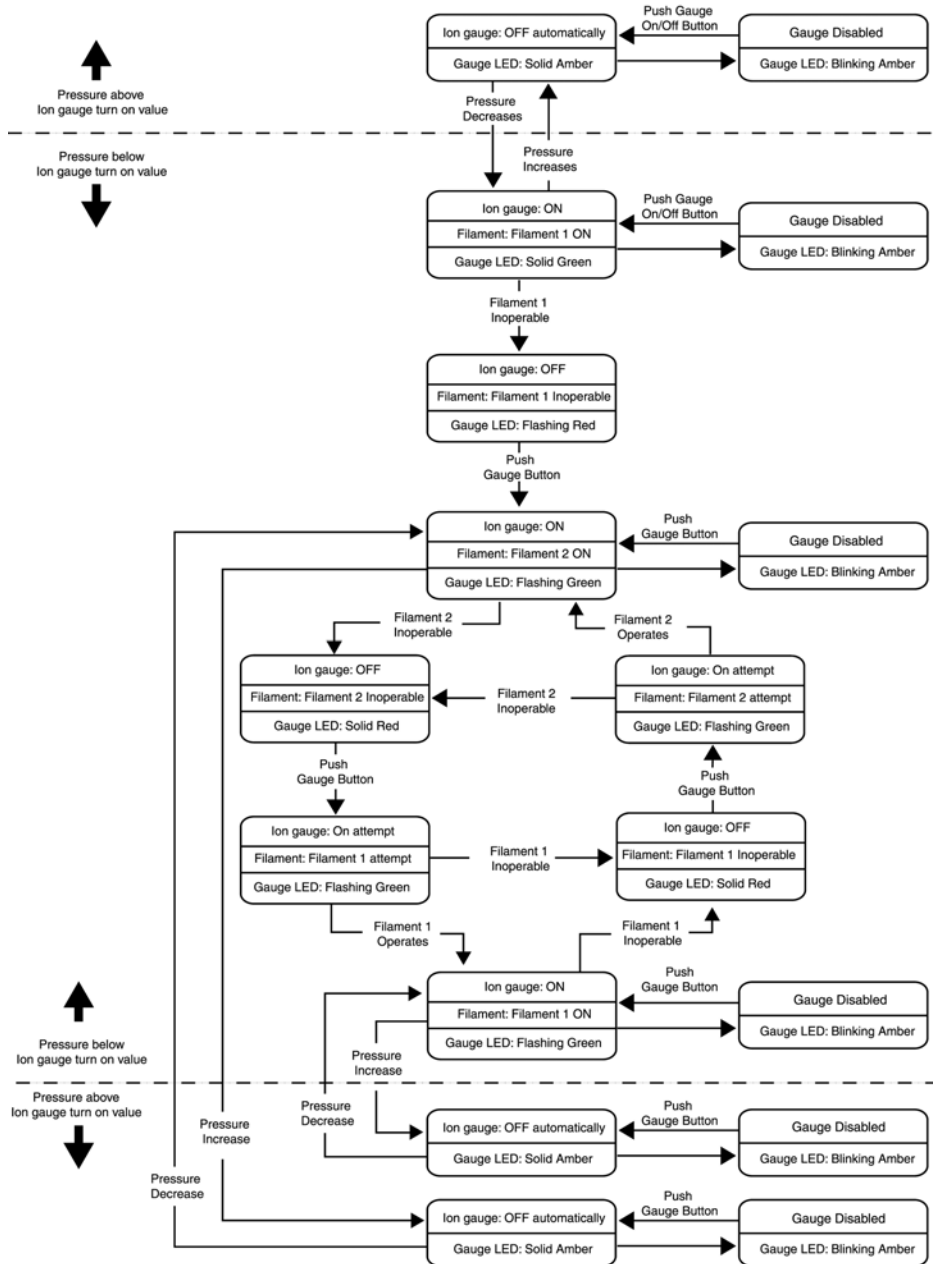


Figure A-3 Manual Filament Switching Flow Chart



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Series 356

Series 356 Micro-Ion[®] Plus Vacuum Gauge Module



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Instruction Manual

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